Graphs and Greedy Algorithm

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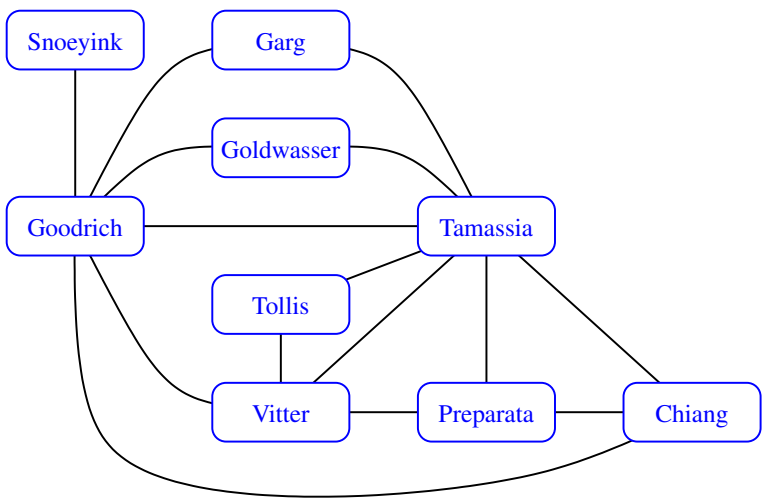
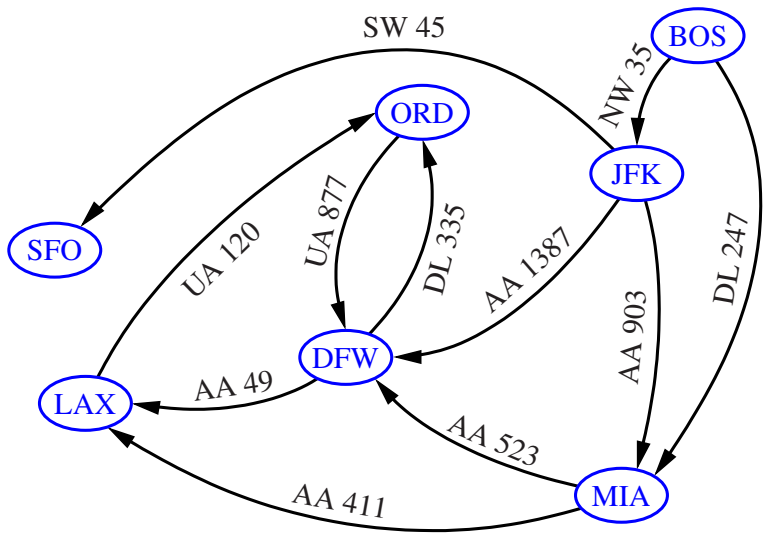
Graph

- A graph is a pair (V, E), where
  - V is a set of nodes, called vertices
  - E is a collection of pairs of vertices, called edges
- Vertices and edges are positions and store elements
- Example:
  - A vertex represents an airport and stores the three-letter airport code
  - An edge represents a flight route between two airports and stores the mileage of the route

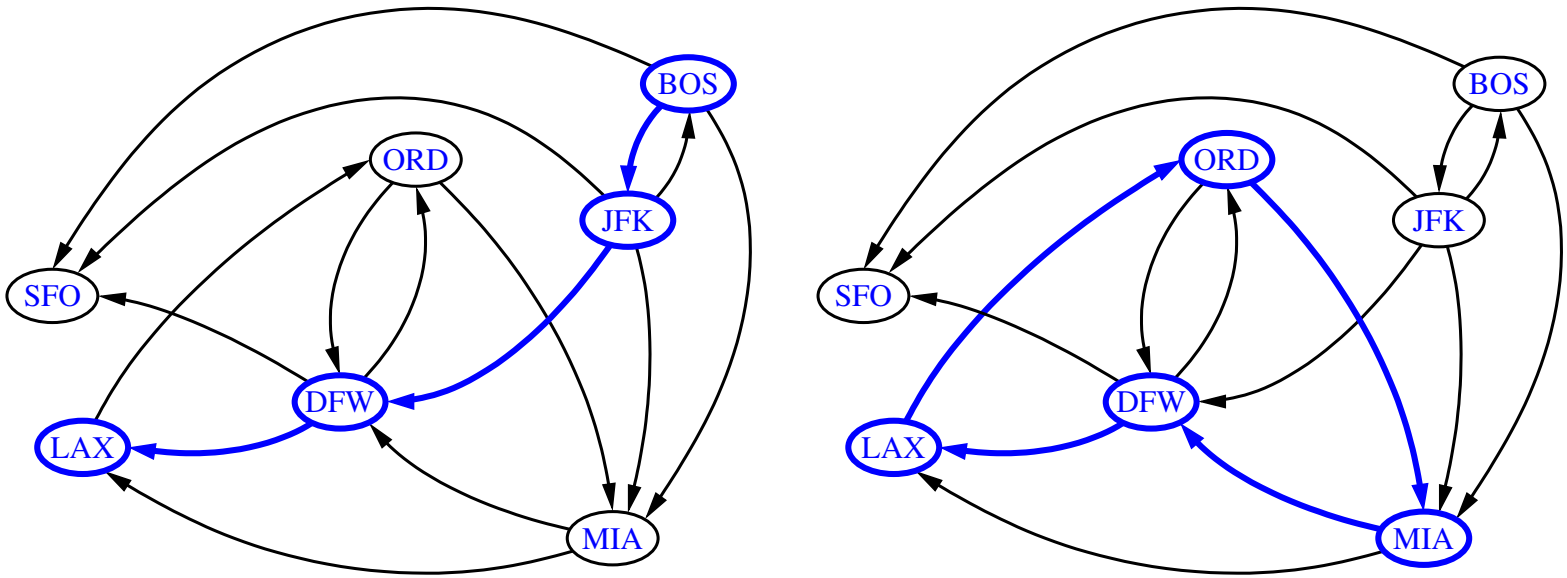




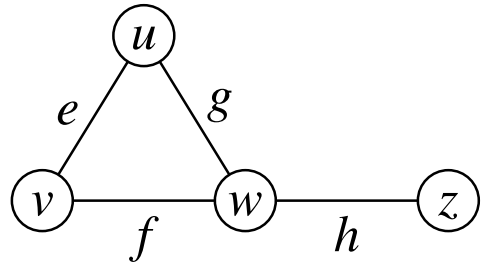
co-author network and flight network

Paths

- Path: a sequence of edges
- Directed path: edges are directed and traversed along the directions
  - $BOS \to JFK \to DFW$: directed simple path
- Cycle: a path that starts and ends at the same vertex
  - $LAX \to ORD \to MIA \to DFW \to LAX$: directed simple cycle



Properties for undirected graphs



Suppose that m is the number of edges, and n is the number of nodes.

Property 1 (Edge and degree)

Let deg(v) denote the degree of v.

$$\sum_{v \in V} deg(v) = 2m \tag{1}$$

Proof Hint: each edge is counted twice

Property 2 (Edge and nodes)

For an undirected graph with no self-loops and no multiple edges

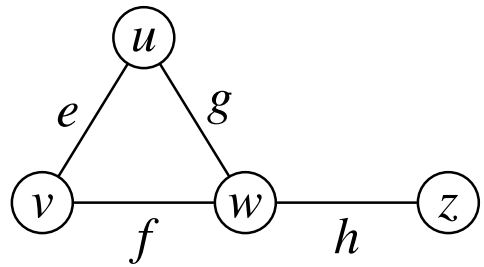
$$m \leq n(n-1)/2 \tag{2}$$

Proof Hint: each vertex has degree at most (n-1).

Complete graph

Complete graph is a graph where every pair of nodes is connected.

Strongly Connected Components (SCC)

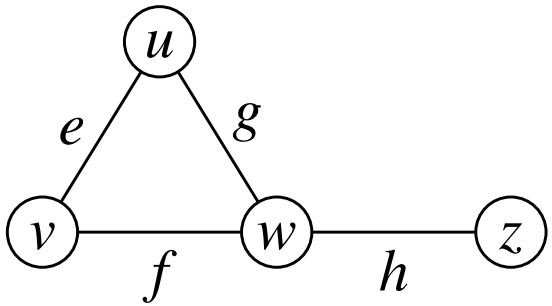


- Node u reaches v: there is a directed path from u to v
- Connected graph: for any two vertices, there is a path between them
- Strongly connected: for any two vertices u and v, u reaches v and v reaches u.
- Strongly connected component(SCC): a subgraph that is strongly connected
- Weakly Connected Component (WCC): a component that, when direction is ignored, every pair of nodes are connected.

How to represent a graph?

- Adjacency matrix
- Edge list
- Adjacency list of edges

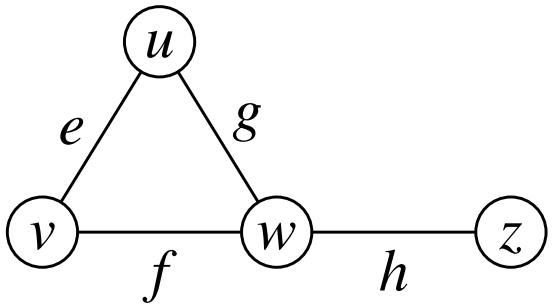
Graph data structures: Adjacency Matrix



		0	1	2	3
u	0		e	g	
v	1	e		f	
w	2	g	f		h
z	3			h	

- Matrix cell (u, v) stores the edge (u, v)
  - if no such edge exists, the slot will store null.
- An intuitive way to represent a graph is matrix

Graph data structures: Adjacency Matrix



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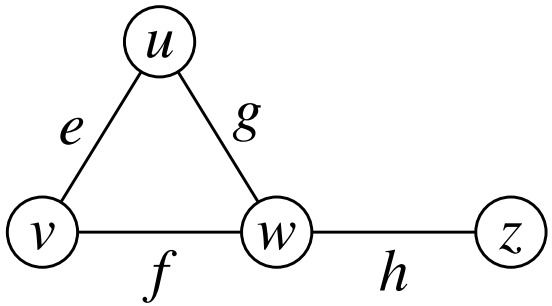

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- Too many empty cells.
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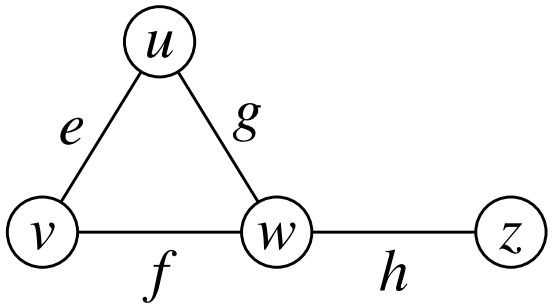

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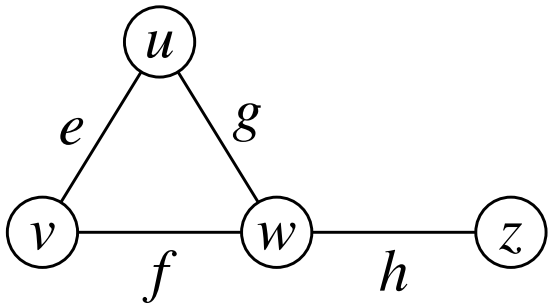
- An intuitive way to represent a graph is matrix
- Problem with matrix representation?
- Too many empty cells.
- The matrix is sparse
  - e.g., if there are 1 billon (10^9) nodes, on average each node has 10 edges
  - There are 10^{18} cells, $10^{18} - 10^{10} \approx 10^{18}$ are empty
  - On average for each useful cell, about 10^8 empty cells are wasted.

Graph data structures: Edge list and Adjacency list

- Graph is represented by a list of edges.

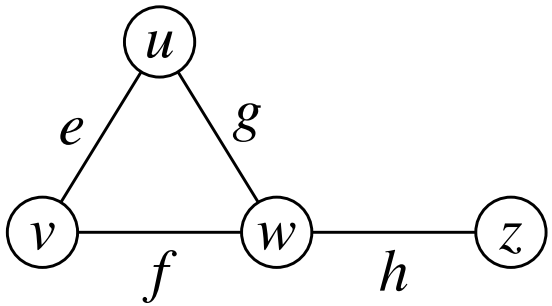


Edge list	
u	v
u	w
v	u
v	w
w	u
w	v
w	z
z	w

		0	1	2	3
u	→ 0		e	g	
v	→ 1	e		f	
w	→ 2	g	f		h
z	→ 3			h	

Graph data structures: Edge list and Adjacency list

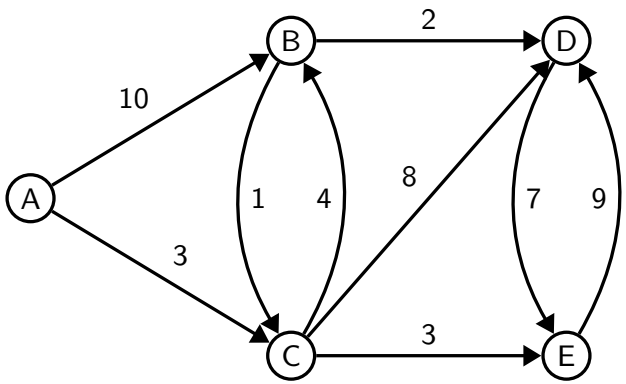
- Graph is represented by a list of edges.
- Can be more efficient if we combine multiple neighbours into one linked list



	0	1	2	3
u → 0		e	g	
v → 1	e		f	
w → 2	g	f		h
z → 3			h	

Edge list		Adjacency list	
u	v		
u	w		
v	u	u	(v, w)
v	w	v	(u, w)
w	u	w	(u, v, z)
w	v	z	(w)
w	z		
z	w		

Graph implementation using edge list



actual text representation

```
5
9
0  1  10
0  2  3
1  2  1
1  3  2
2  1  4
2  3  8
2  4  3
3  4  7
4  3  9
```

- Line one: number of nodes
- Line two: number of edges
- Each line afterwards: edge (from, to, weight)
- Use positive integers to denote nodes

Graph class

The constructor reads from a text file that keeps edges of a graph (and the weights in this example)

```
public class Graph {
        public int V;
        public int E;
        //Set of edges for each node
        public Set<Edge>[] edgesOf;
        public int[] indegree;

        public Graph(Scanner in) throws Exception {
                V = in.nextInt();
                E = in.nextInt();
                indegree = new int[V];
                edgesOf = (Set<Edge>[]) new HashSet[V];
                for (int v = 0; v < V; v++)
                        edgesOf[v] = new HashSet<Edge>();
                for (int i = 0; i < E; i++) {
                        int v = in.nextInt();
                        int w = in.nextInt();
                        double weight = in.nextDouble();
                        edgesOf[v].add(new Edge(v,w,weight));
                        indegree[w]++;
                }
        }
```

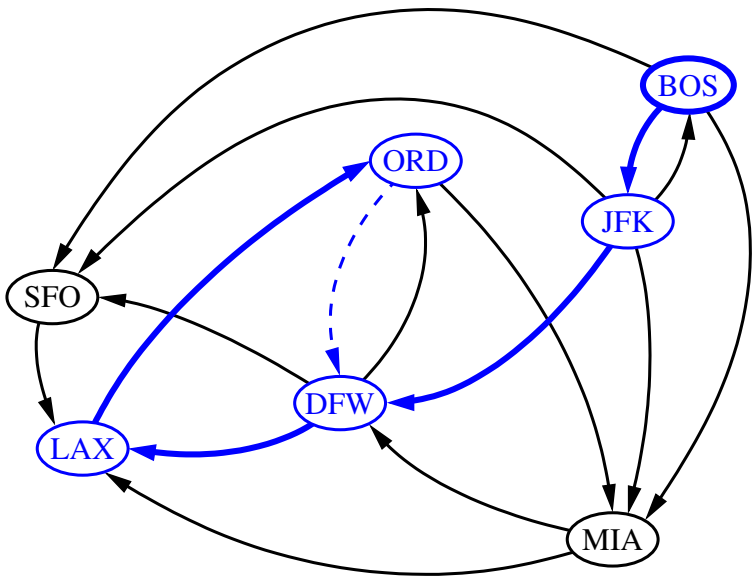
```
public class Edge {
    private final int v;
    private final int w;
    private final double weight;

    public Edge(int v, int w, double weight) {
        this.v = v;
        this.w = w;
        this.weight = weight;
    }

    public int from() {
        return v;
    }

    public int to() {
        return w;
    }

     public double weight() {
        return weight;
    }

    public String toString() {
        return "(" + v +" "+ String.format("%5.2f", weight)+"
             "+w+")";
    }
 }
```

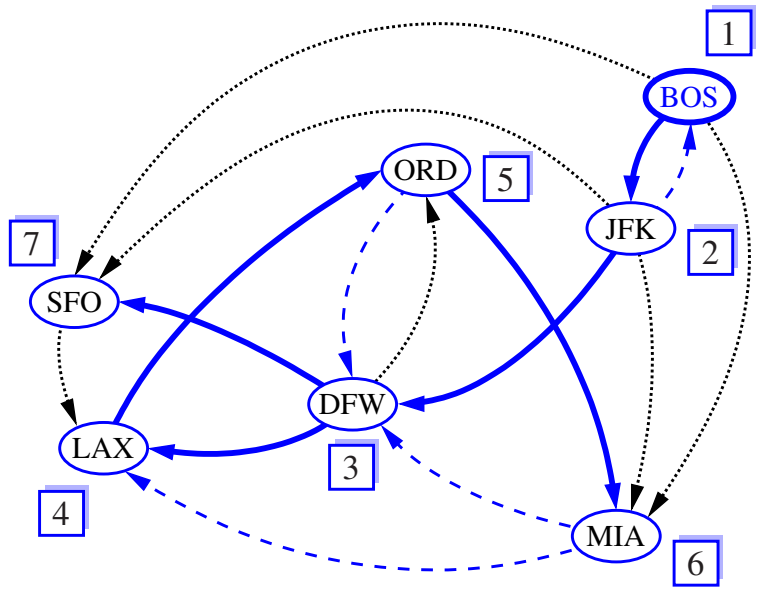
Graph traversals

- Traversal: a systematic procedure for exploring a graph by examining all its vertices
- Two typical approaches:
  - DFS
  - BFS

Depth First Search

DFS starting from BOS

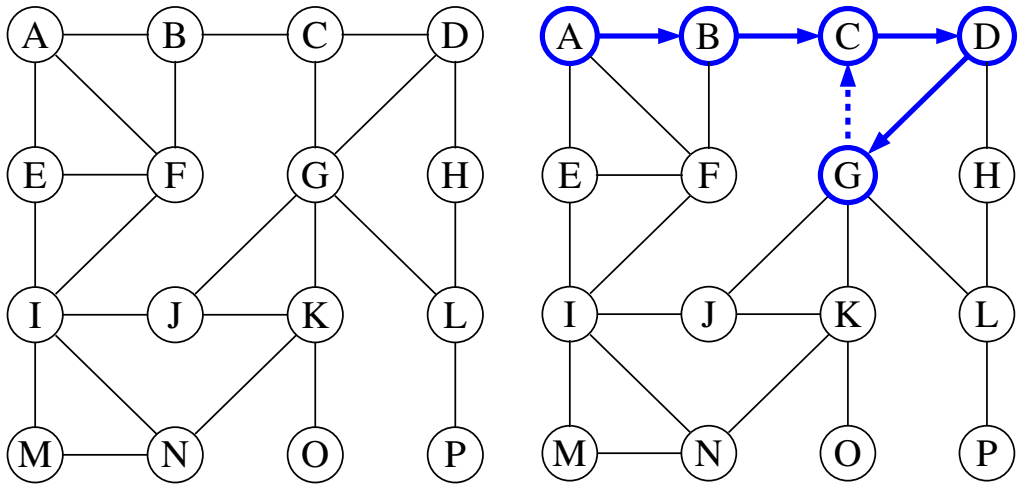




Back edge

Tree edge (discovery edge)

Depth First Search: Algorithm

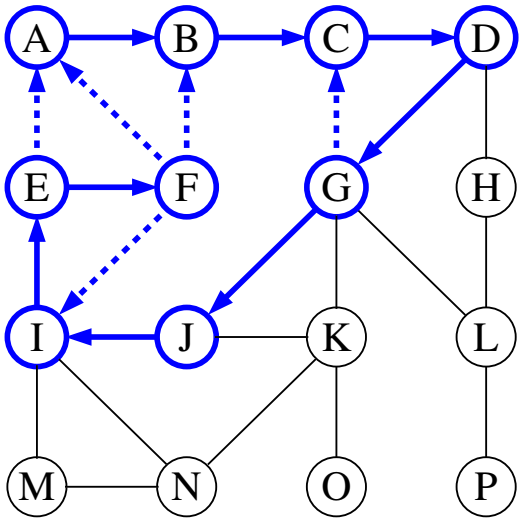


A→ B → C → D→G → C?

Algorithm 1: DFS(G, u)

```
1 record(u);
2 for (u,v) ∈ G do
3 |   if v was not visited then
4 |   |   DFS(G,v);
5 |   end
6 end
```

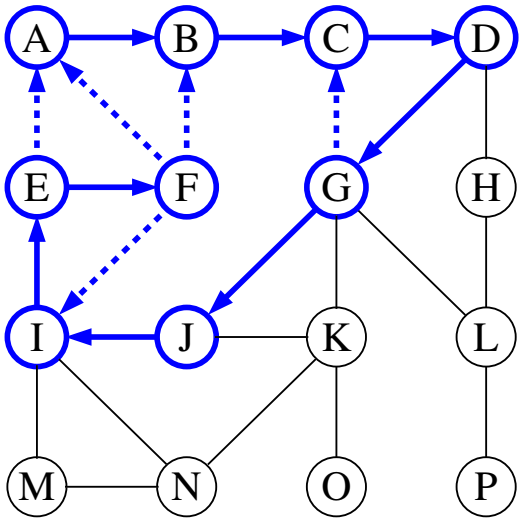
DFS




```
private static void dfs(Graph G, int v) {
        marked[v] = true;
        for (Edge e : G.edgesOf[v]) {
                int adj = e.to();
                if (!marked[adj])
                        dfs(G, adj);
        }
}
```

G→J

DFS

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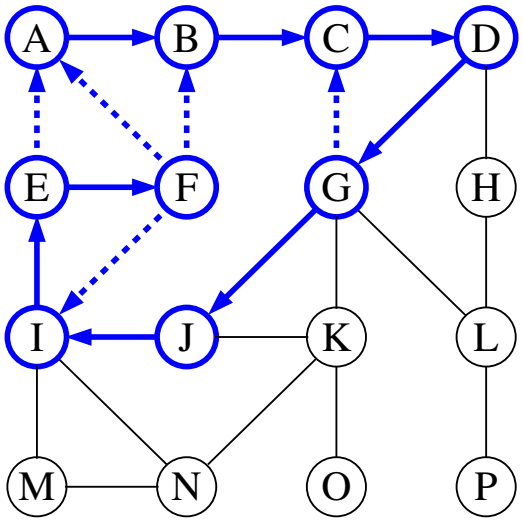
G→J →I

DFS

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        }
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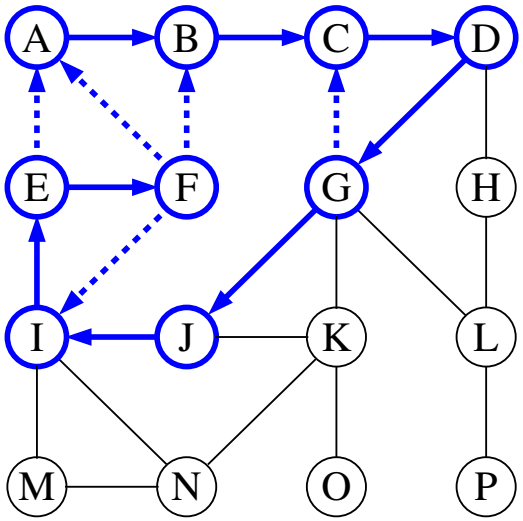
```

G→J →I → E

DFS

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}
```
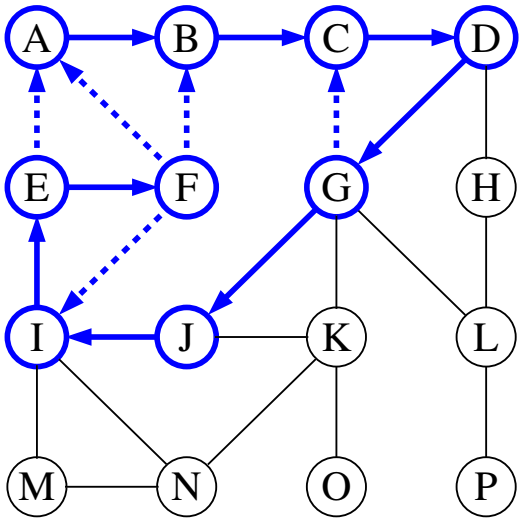

G→J →I → E → A?

DFS

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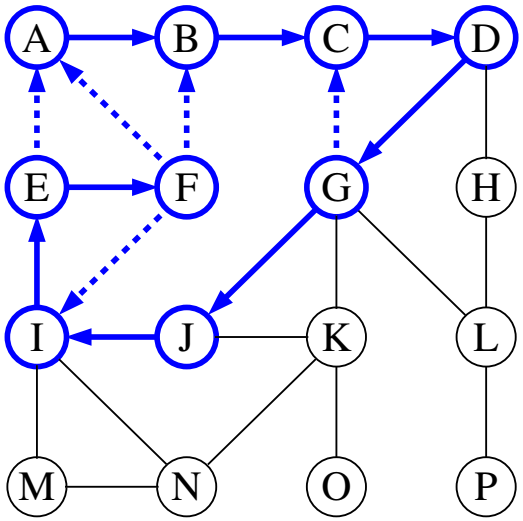
G→J →I → E → A? → F

DFS

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        }
}
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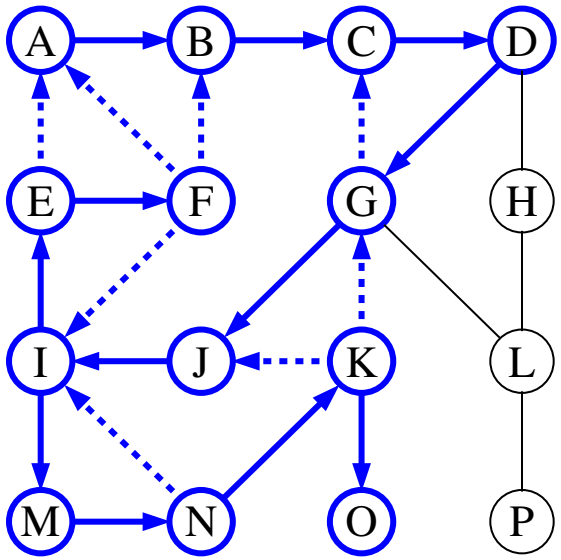
G→J →I → E → A? → F → B?

DFS



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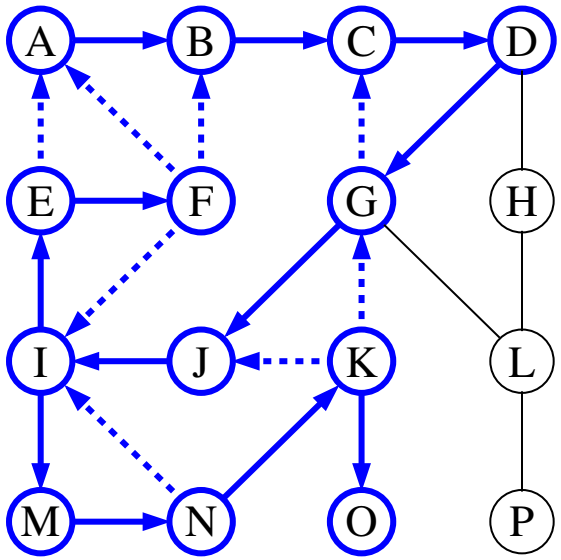
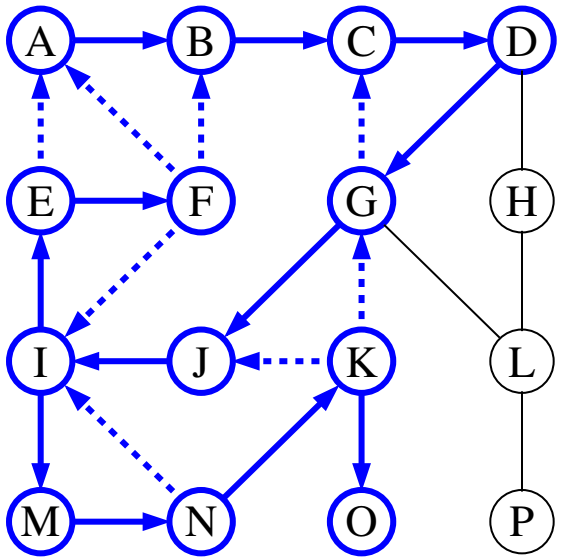


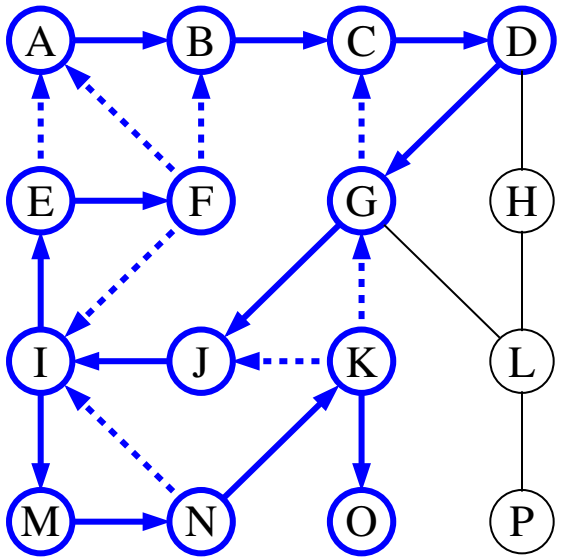
I $\rightarrow$ M

DFS



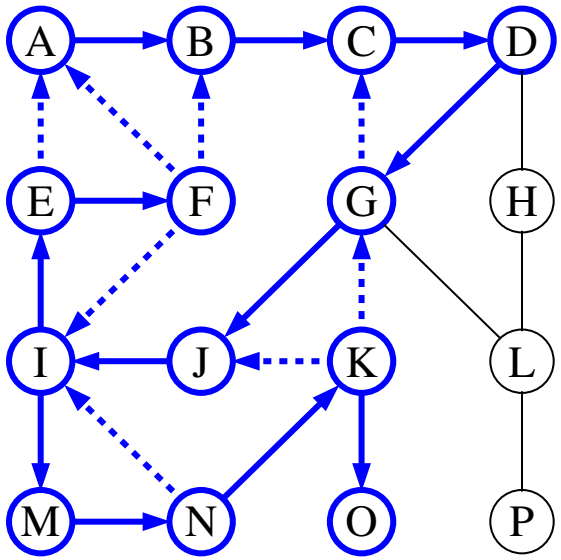
$I \rightarrow M \rightarrow N$

DFS



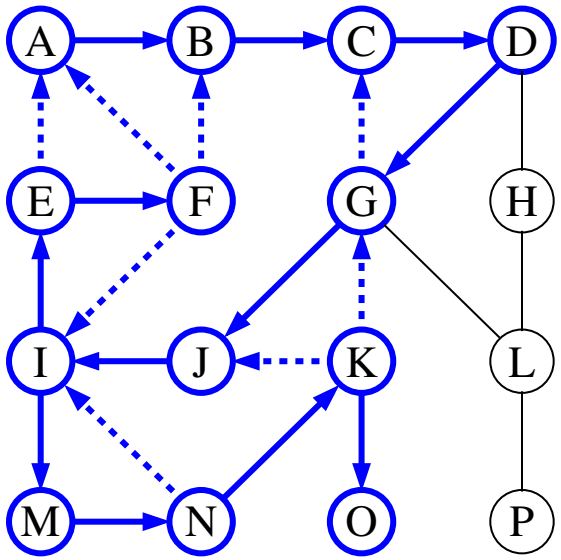
$I \rightarrow M \rightarrow N \rightarrow I$?

DFS



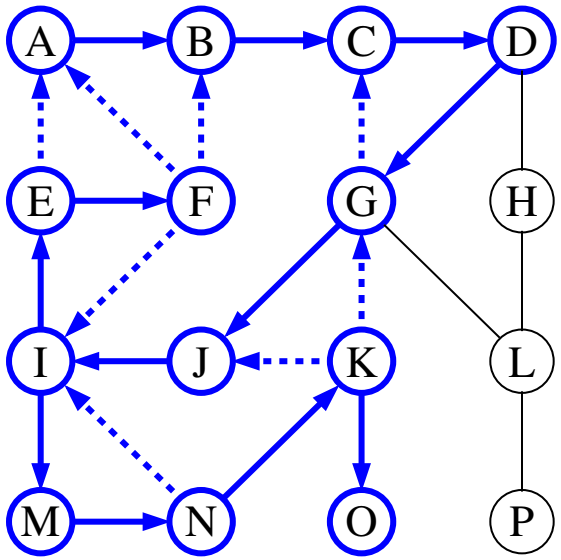
$I \rightarrow M \rightarrow N \rightarrow I\ ? \rightarrow K$

DFS



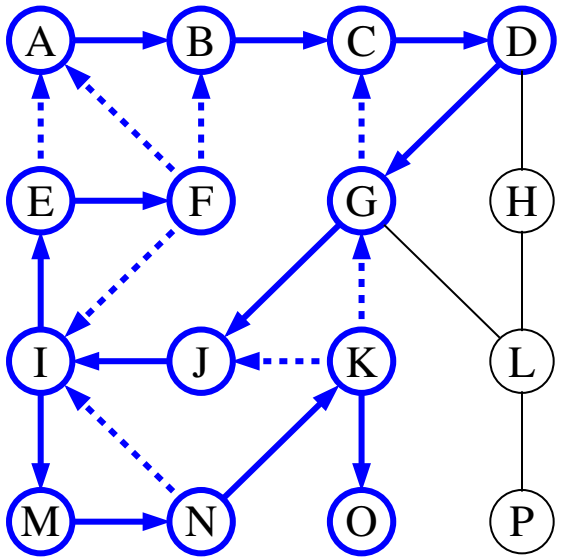
I $\rightarrow$ M $\rightarrow$ N $\rightarrow$ I ? $\rightarrow$ K $\rightarrow$ J?

DFS



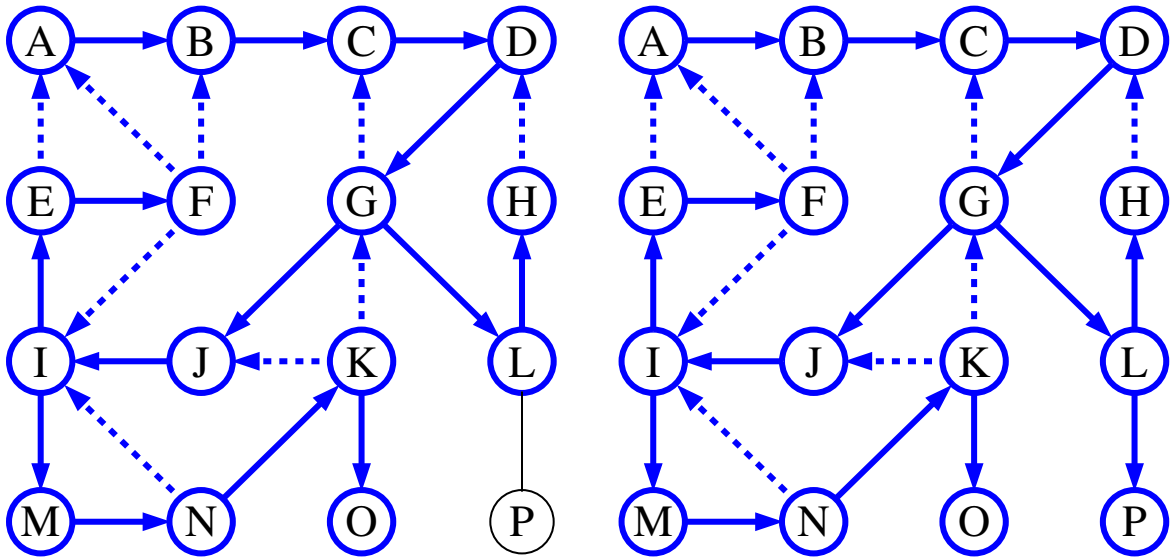
I $\rightarrow$ M $\rightarrow$ N $\rightarrow$ I ? $\rightarrow$ K $\rightarrow$ J? $\rightarrow$ G?

DFS



I $\rightarrow$ M $\rightarrow$ N $\rightarrow$ I ? $\rightarrow$ K $\rightarrow$ J? $\rightarrow$ G? $\rightarrow$ O

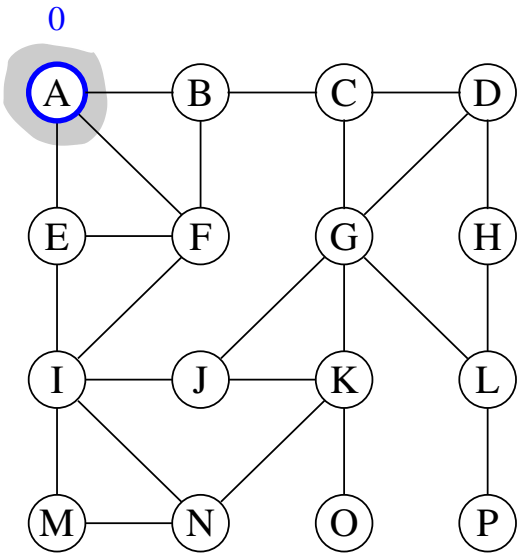
DFS



BFS (Breadth first search)

Applications

- Web crawling
- Social networks
- ....



Q=(a)

BFS (Breadth first search)

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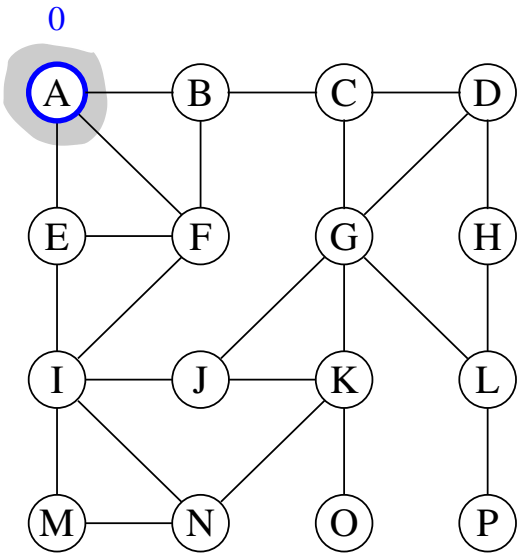

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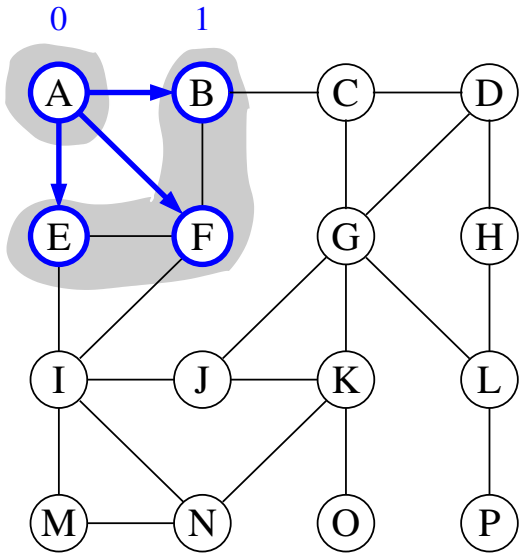

Q=(E,F,B)

BFS (Breadth first search)

Applications

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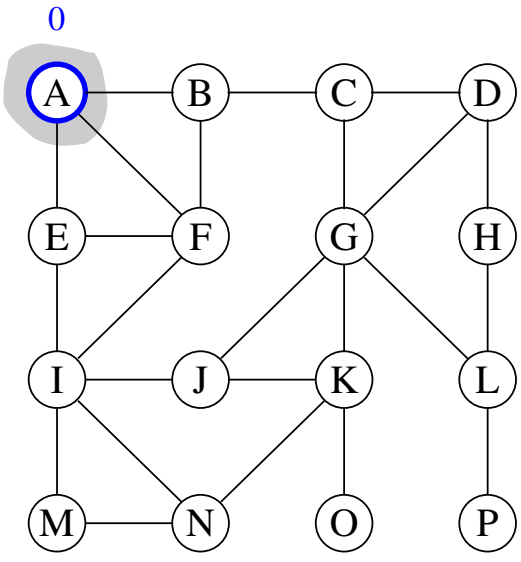

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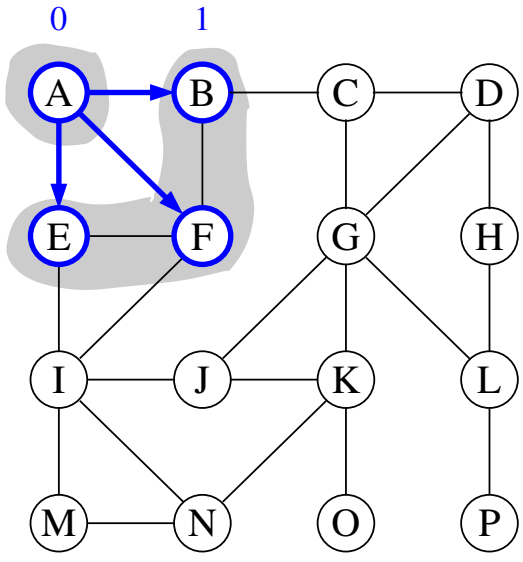
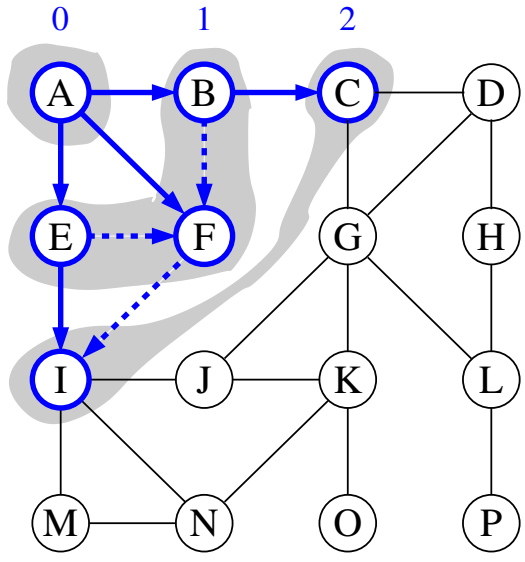


Q=(E,F,B)

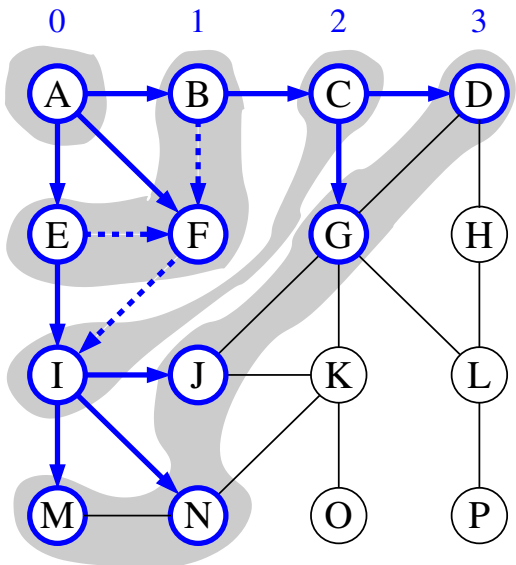


Q=(I,C)

BFS

Algorithm 2: BFS(G, u)

```
1 add u to queue Q.
2 while Q not empty do
3 |   v=Q.dequeue();
4 |   record(v)
5 |   add all unmarked neighbour of v to Q.
6 |   mark those neighbiours
7 end
```

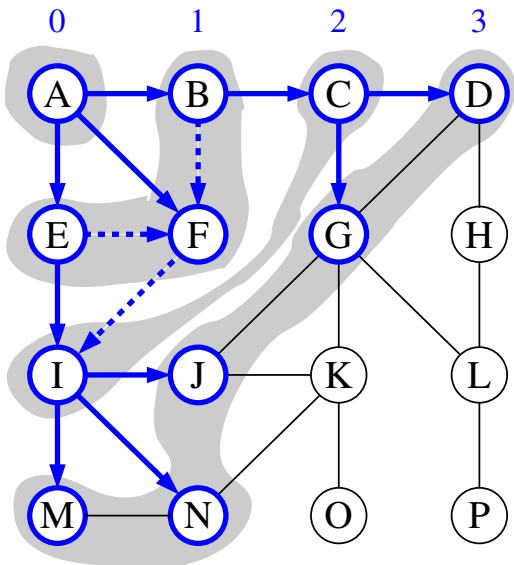


Q=(M,N,J,G,D)

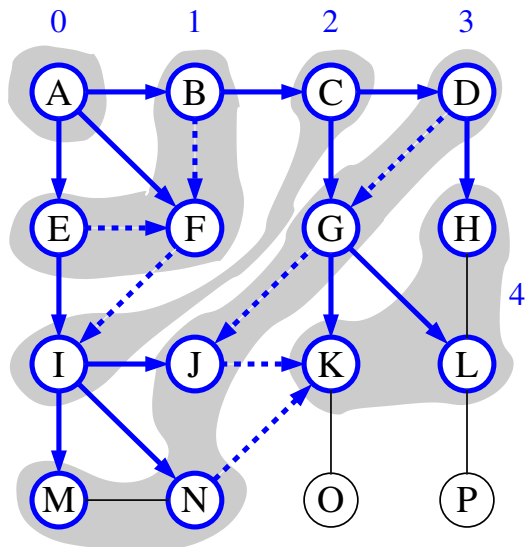
BFS

Algorithm 3: BFS(G, u)

```
1 add u to queue Q.
2 while Q not empty do
3 |   v=Q.dequeue();
4 |   record(v)
5 |   add all unmarked neighbour of v to Q.
6 |   mark those neighbiours
7 end
```



Q=(M,N,J,G,D)

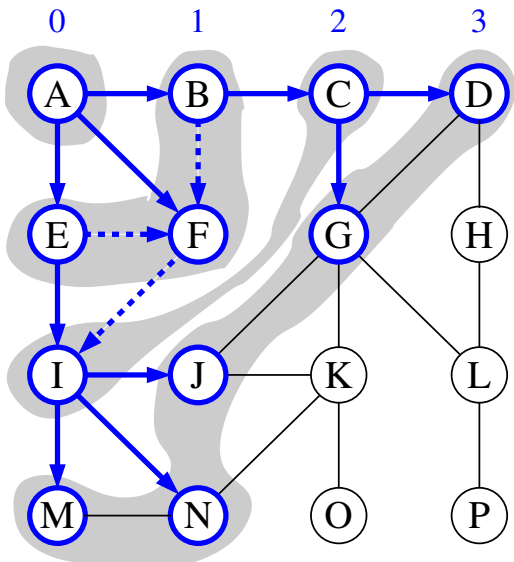


Q=(K,L,H)

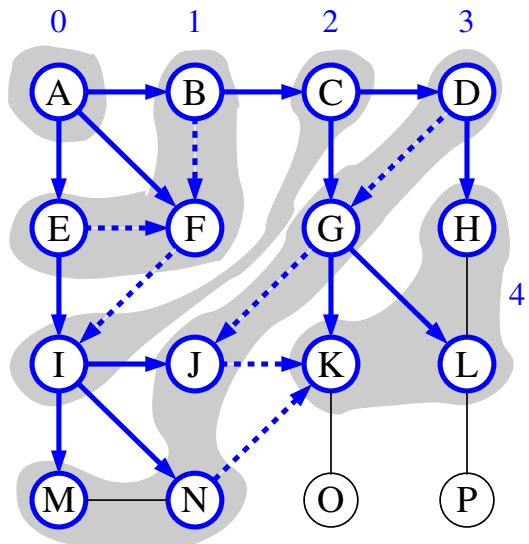
BFS

Algorithm 4: BFS(G, u)

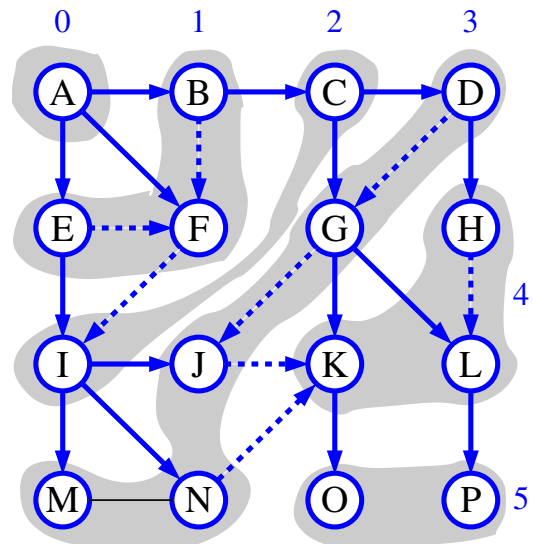
```
1 add u to queue Q.
2 while Q not empty do
3 |   v=Q.dequeue();
4 |   record(v)
5 |   add all unmarked neighbour of v to Q.
6 |   mark those neighbiours
7 end
```



Q=(M,N,J,G,D)



Q=(K,L,H)



Q=(O,P)

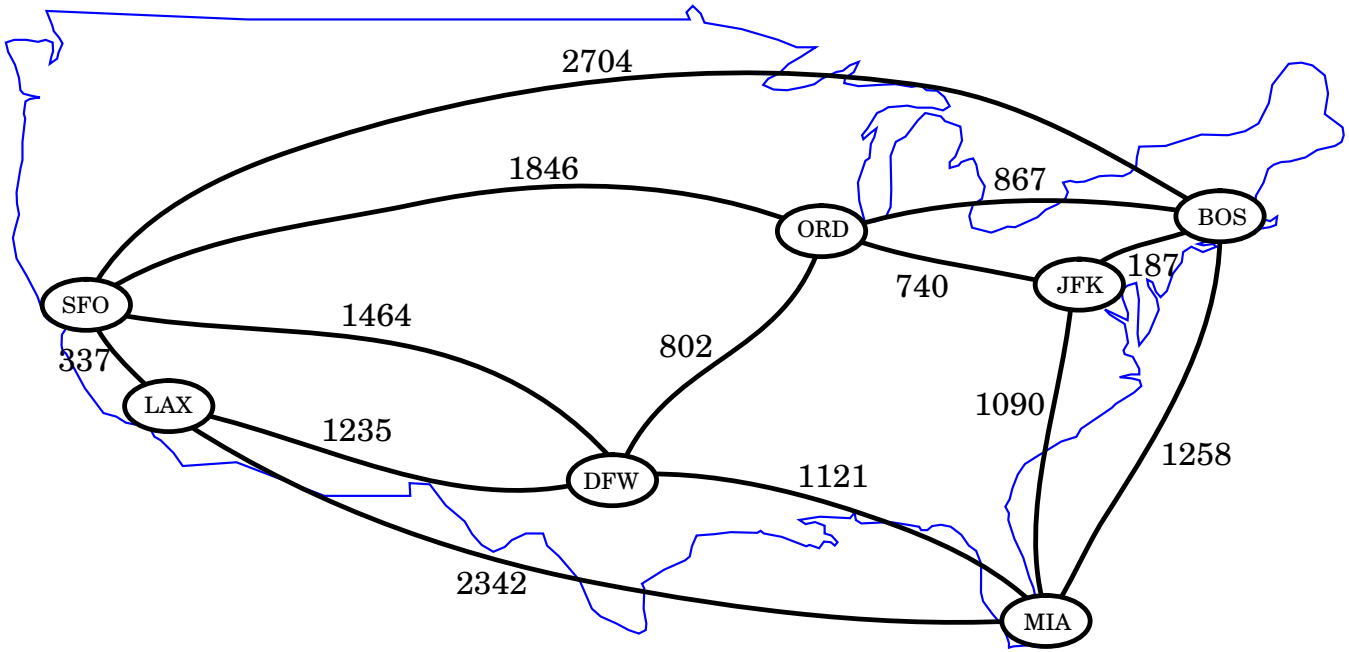
BFS Code

```
public static Queue<Integer> BFS(Graph G, int s) {
        Queue<Integer> visited = new LinkedList<Integer>();
        boolean[]       marked = new boolean[G.V]; // node in queue
        int[] distTo = new int[G.V]; // distance to the source s
        int[] edgeTo=new int[G.V];//previous edge on shortest path
        Queue<Integer> q = new LinkedList<Integer>();
        for (int v = 0; v < G.V; v++)
                distTo[v] = Integer.MAX_VALUE;
        distTo[s] = 0;
        q.add(s);
        marked[s] = true;

        while (!q.isEmpty()) {
                int v = q.remove();
                visited.add(v);
                for (Edge e : G.edgesOf[v]) {
                        int w = e.to(); //w is a neighbour
                        if (!marked[w]) {
                                edgeTo[w] = v;
                                distTo[w] = distTo[v] + 1;
                                q.add(w);
                                marked[w] = true;
                        }
                }
        }
        return visited;
}
```

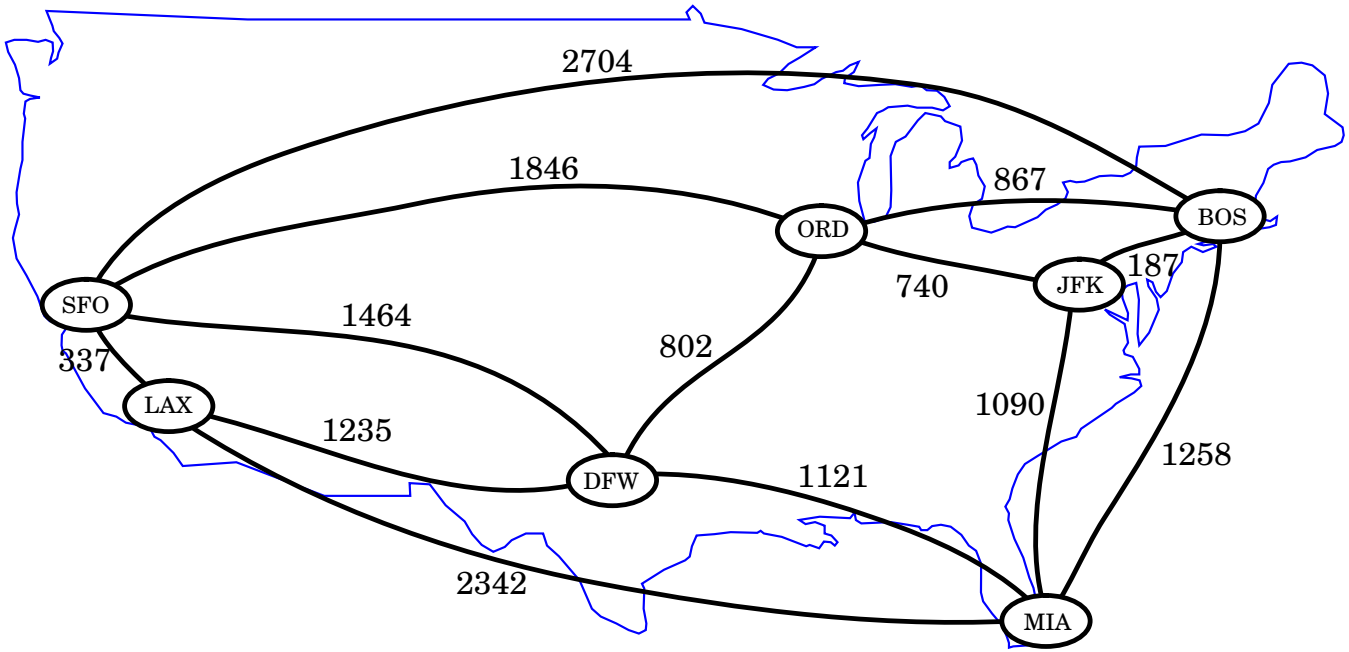
It is a shortest path algorithm for unweighted graph.

Shortest path (for weighted graph)



- Weighted graph: edges have weight
- E.g.: the weight is the distance between airports.
- What is the time complexity of a bad algorithm?

Shortest path (for weighted graph)



- Weighted graph: edges have weight
- E.g.: the weight is the distance between airports.
- What is the time complexity of a bad algorithm?
- It can be exponential

Shortest path and Dijkstra's algorithm: problem definition

There are variants of shortest path problem and Dijkstra algorithms

- Graphs
  - Weighted/unweighted graph
  - Directed/undirected graphs
  - Whether weight can be negative
- Solutions
  - *(s,t) shortest path*: find the shortest path from node s to node t.
  - *single-source shortest path*: from a source node s, find shortest paths to all other nodes.
- Output
  - The shortest Path
  - The path length only (the original algorithm)
- Implementations
  - Unsorted array
  - Hash?
  - Priority queue(Heap) ?

Greedy algorithm

- Greedy algorithm is an algorithm design paradigm
- In each stage make the local optimal choice
- Often used in optimization problems
- Dijkstra's algorithm is an example of greedy algorithms
- Compare with other paradigms
  - Divide and Conquer (e.g., merge sort, quick Sort)
  - Dynamic programming (e.g., Fibonacci, edit distance)

Dijkstra's algorithm

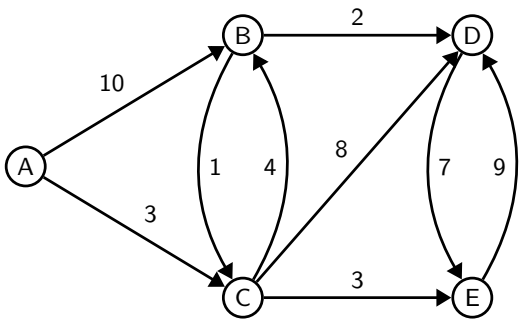
- Implement Dijkstra's algorithm with Priority Queue.
- Question: how to update Q efficiently?

Algorithm 5: Dijkstra's algorithm

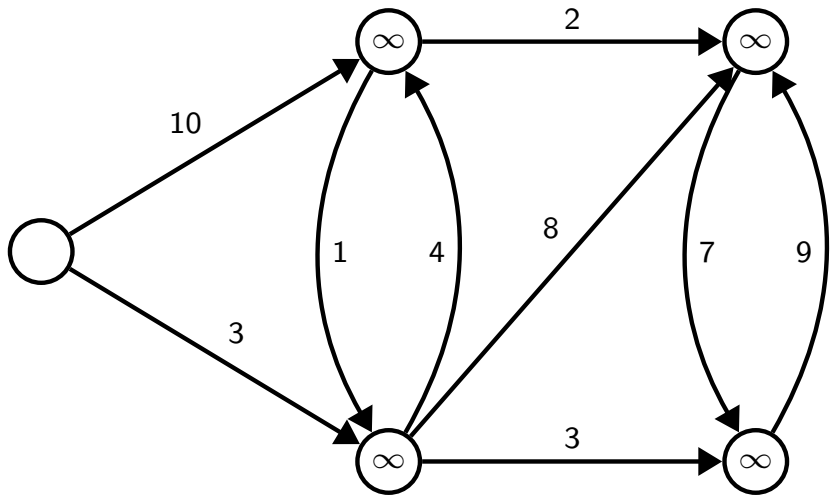
Input: Graph G=(V, E), starting node *s*
Output: Distance *dist* to all remaining nodes

```
1  dist[s] =0;
2  dist[v]=∞ for all other nodes;
3  PriorityQueue Q=V with their distances;
4  while Q not empty do
5  |   u =Q.removeMin();
6  |   for all v ∈ neighbors[u] and v ∈ Q do
7  |   |   if dist[v] > dist[u] + w(u, v) then
8  |   |   |   dist[v]=dist[u] + w(u, v);
9  |   |   |   update Q with new distance
10 |   |   end
11 |   end
12 end
13 return dist
```

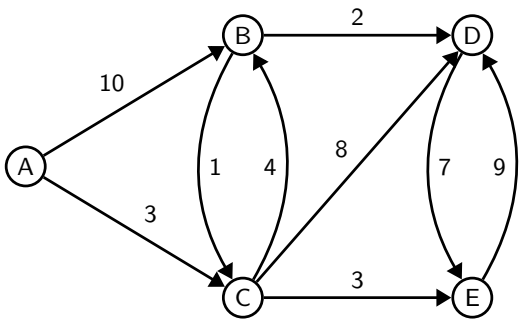
Tracing the algorithm



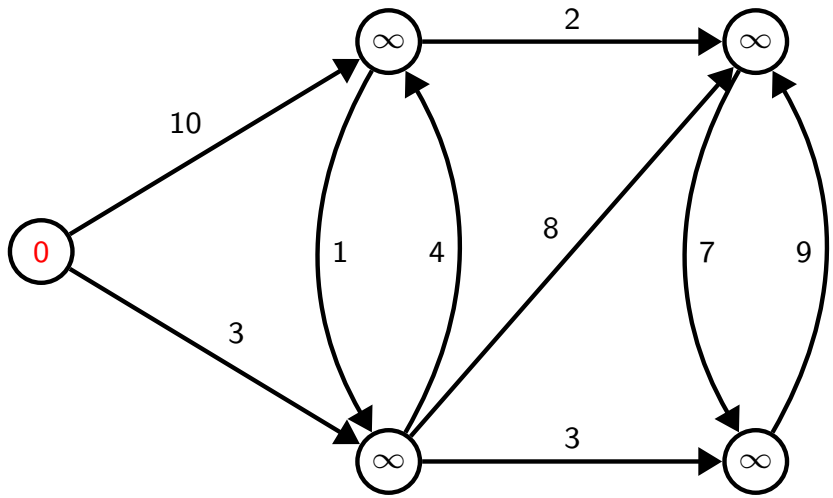
	A	B	C	D	E	Q	u
0	0	∞	∞	∞	∞	A,B,C,D,E	A



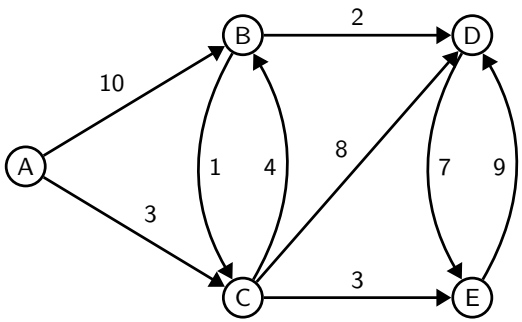
Tracing the algorithm



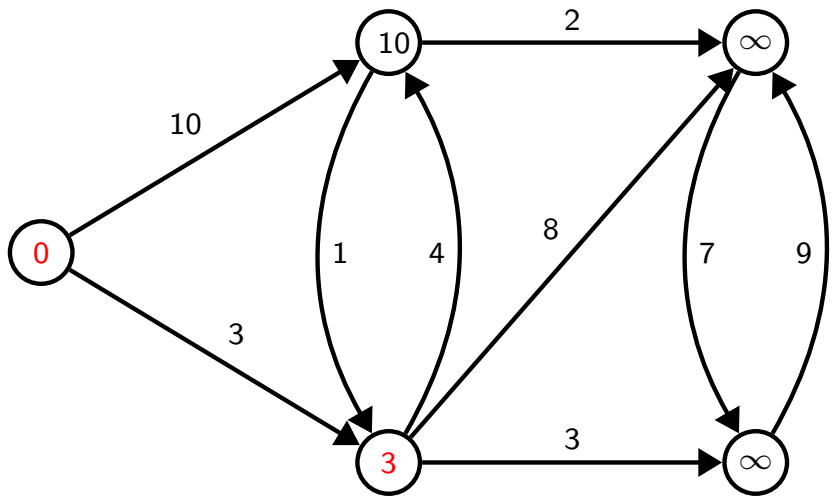
	A	B	C	D	E	Q	u
0	0	∞	∞	∞	∞	A,B,C,D,E	A
1	0	10	3	∞	∞	B,C,D,E	C



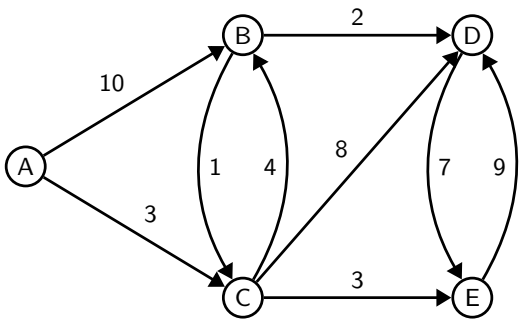
Tracing the algorithm



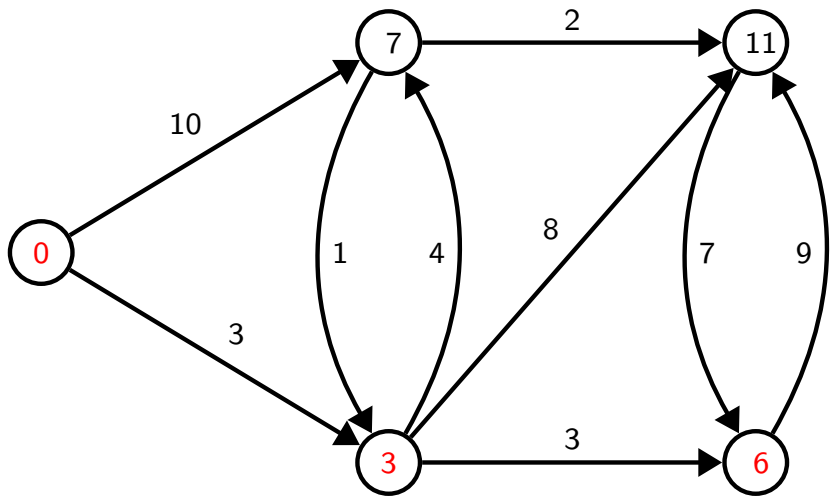
	A	B	C	D	E	Q	u
0	0	∞	∞	∞	∞	A,B,C,D,E	A
1	0	10	3	∞	∞	B,C,D,E	C
2	0	7	3	11	6	B,D,E	E



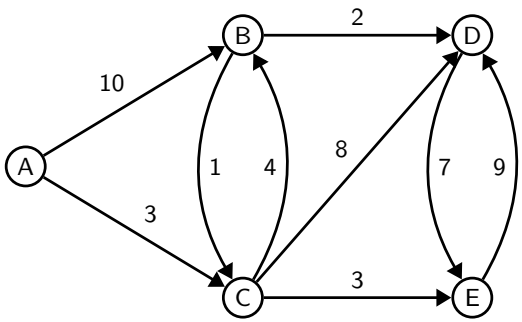
Tracing the algorithm



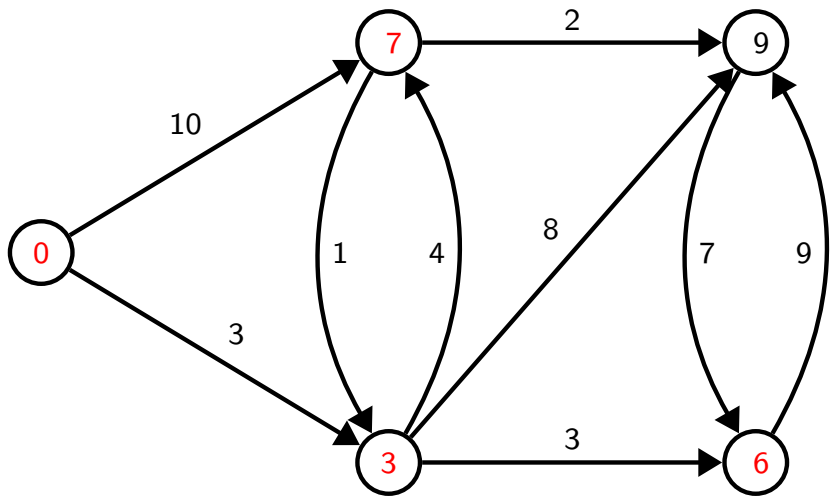
	A	B	C	D	E	Q	u
0	0	∞	∞	∞	∞	A,B,C,D,E	A
1	0	10	3	∞	∞	B,C,D,E	C
2	0	7	3	11	6	B,D,E	E
3	0	7	3	11	6	B,D	B



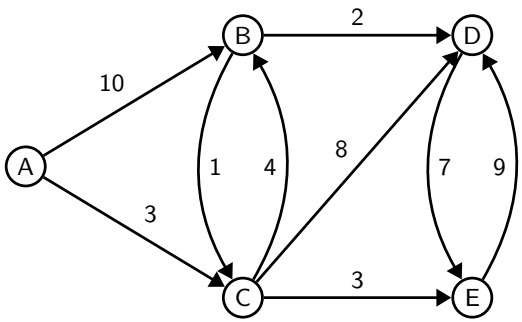
Tracing the algorithm



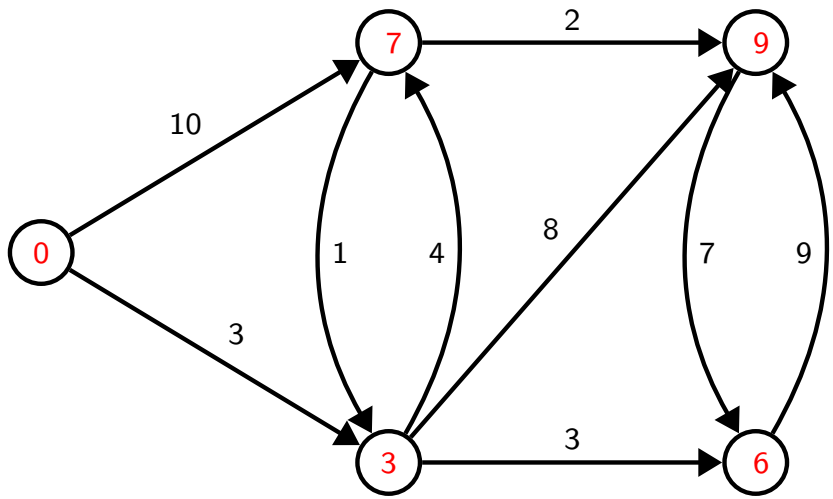
	A	B	C	D	E	Q	u
0	0	∞	∞	∞	∞	A,B,C,D,E	A
1	0	10	3	∞	∞	B,C,D,E	C
2	0	7	3	11	6	B,D,E	E
3	0	7	3	11	6	B,D	B
4	0	7	3	9	6	D	D



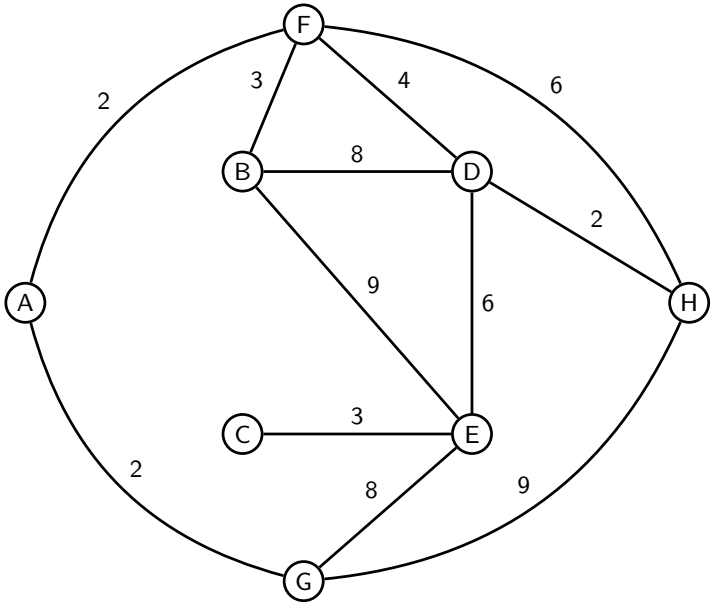
Tracing the algorithm

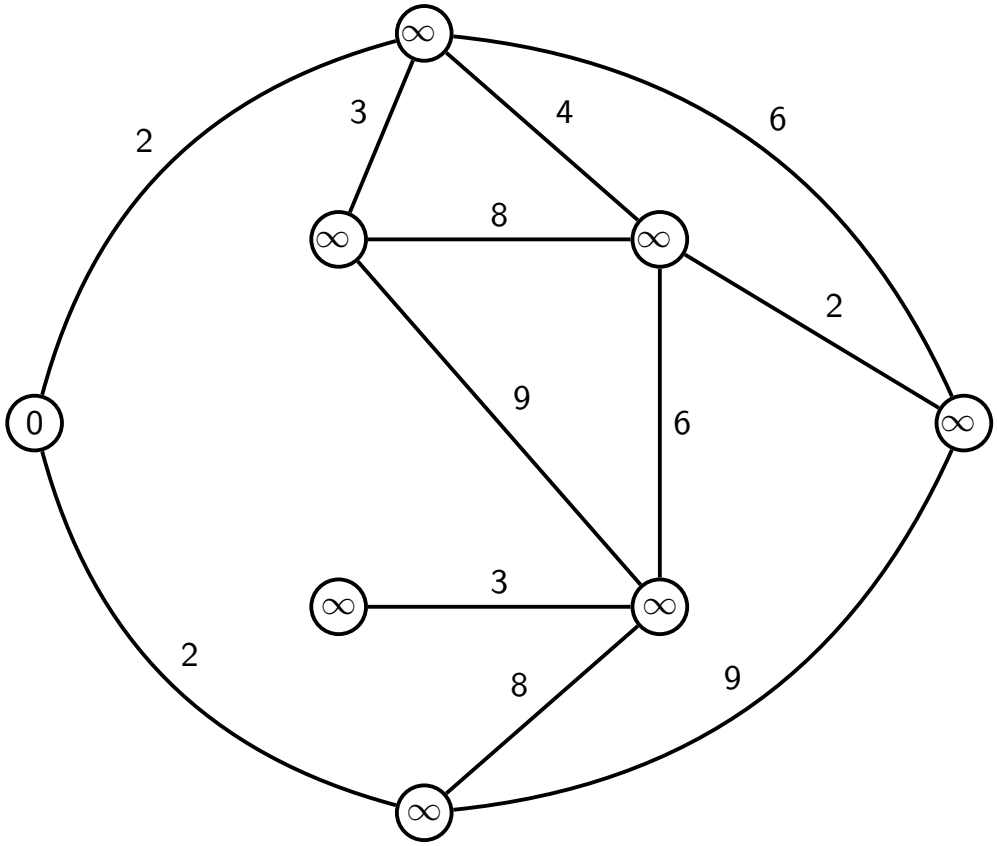


	A	B	C	D	E	Q	u
0	0	∞	∞	∞	∞	A,B,C,D,E	A
1	0	10	3	∞	∞	B,C,D,E	C
2	0	7	3	11	6	B,D,E	E
3	0	7	3	11	6	B,D	B
4	0	7	3	9	6	D	D
5	0	7	3	9	6		



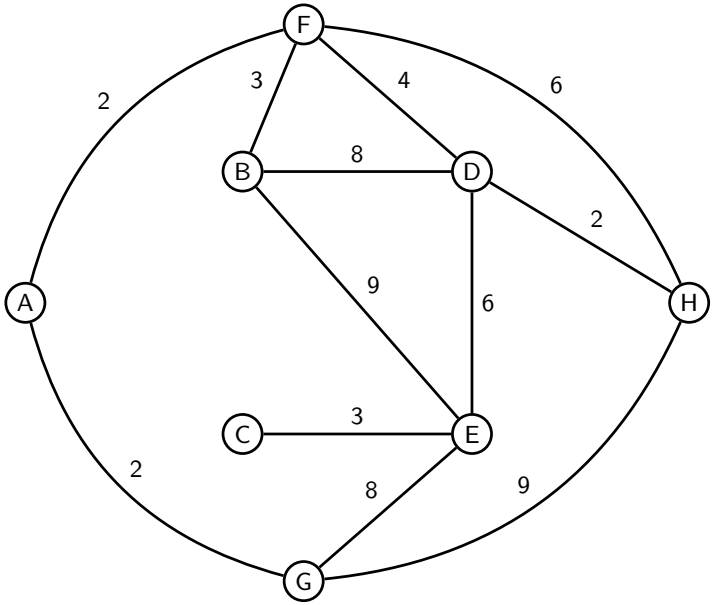
Tracing the algorithm: second example

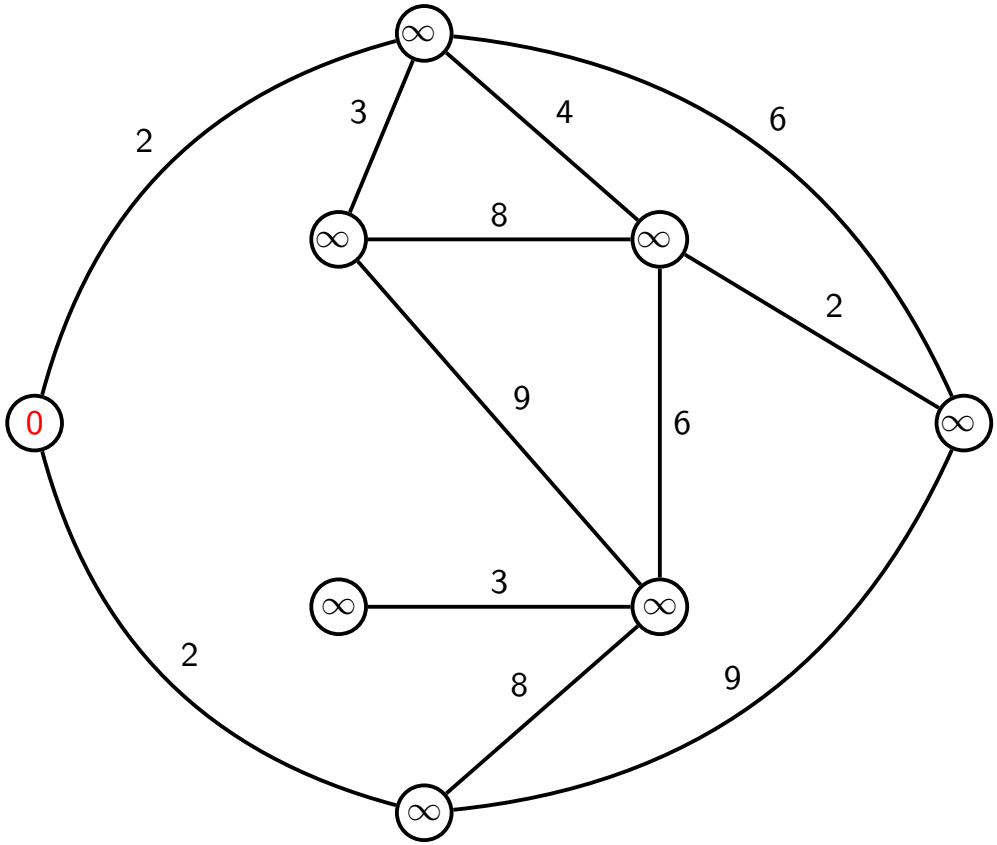




A	B	C	D	E	F	G	H	Q	u

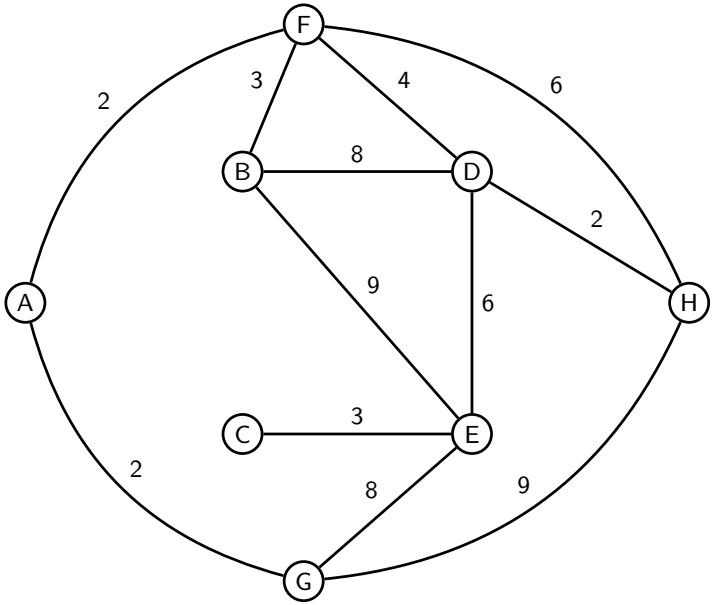
Tracing the algorithm: second example

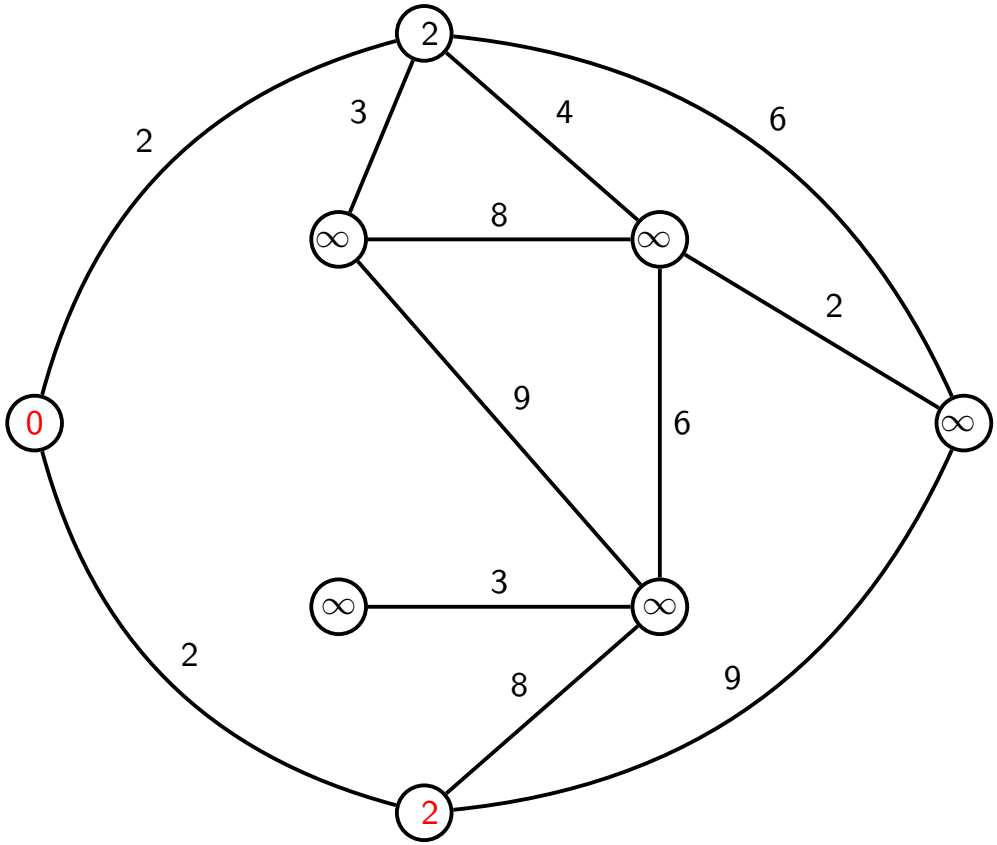




	A	B	C	D	E	F	G	H	Q	u
0	0	∞	∞	∞	∞	∞	∞	∞	A,B,C,D,E,F,G,H	A

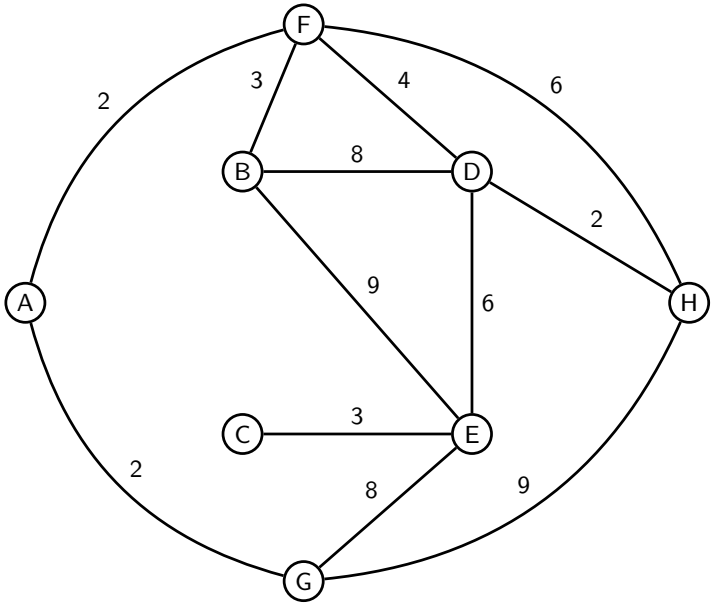
Tracing the algorithm: second example

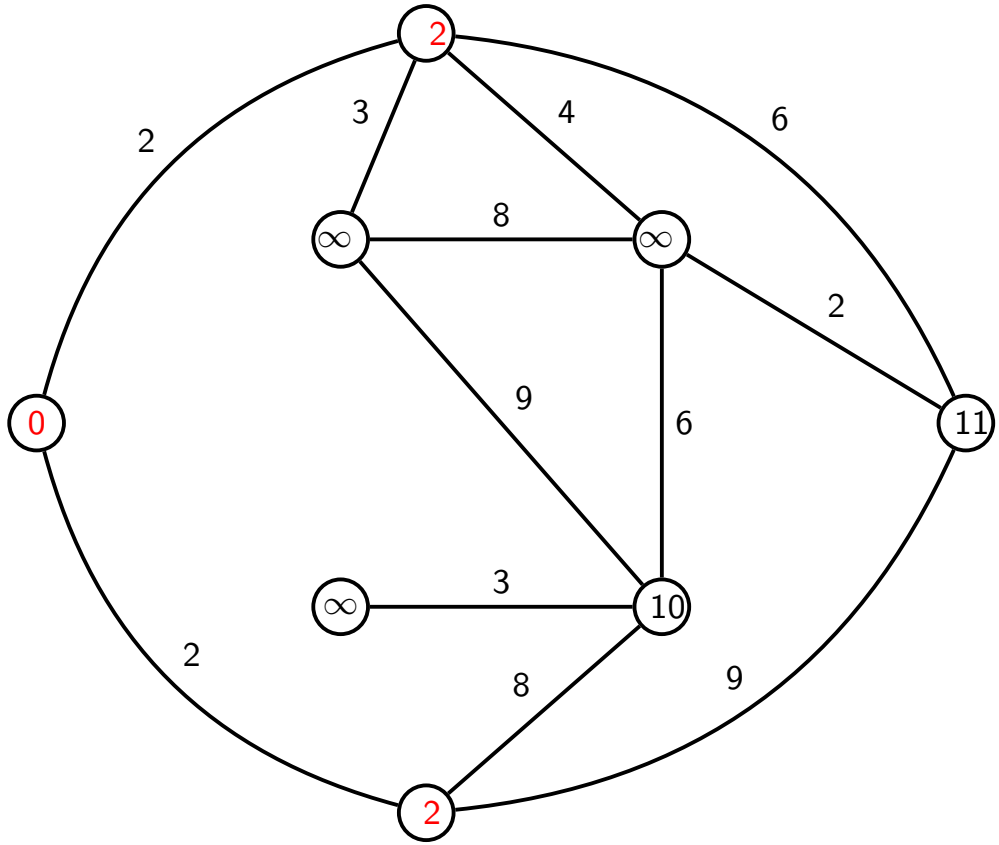




	A	B	C	D	E	F	G	H	Q	u
0	0	∞	∞	∞	∞	∞	∞	∞	A,B,C,D,E,F,G,H	A
1	0	∞	∞	∞	∞	2	2	∞	B,C,D,E,F,G,H	G

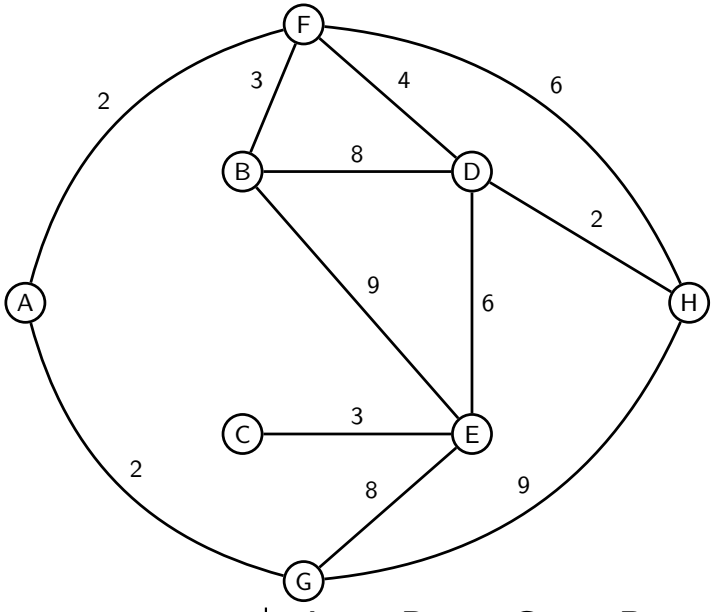
Tracing the algorithm: second example

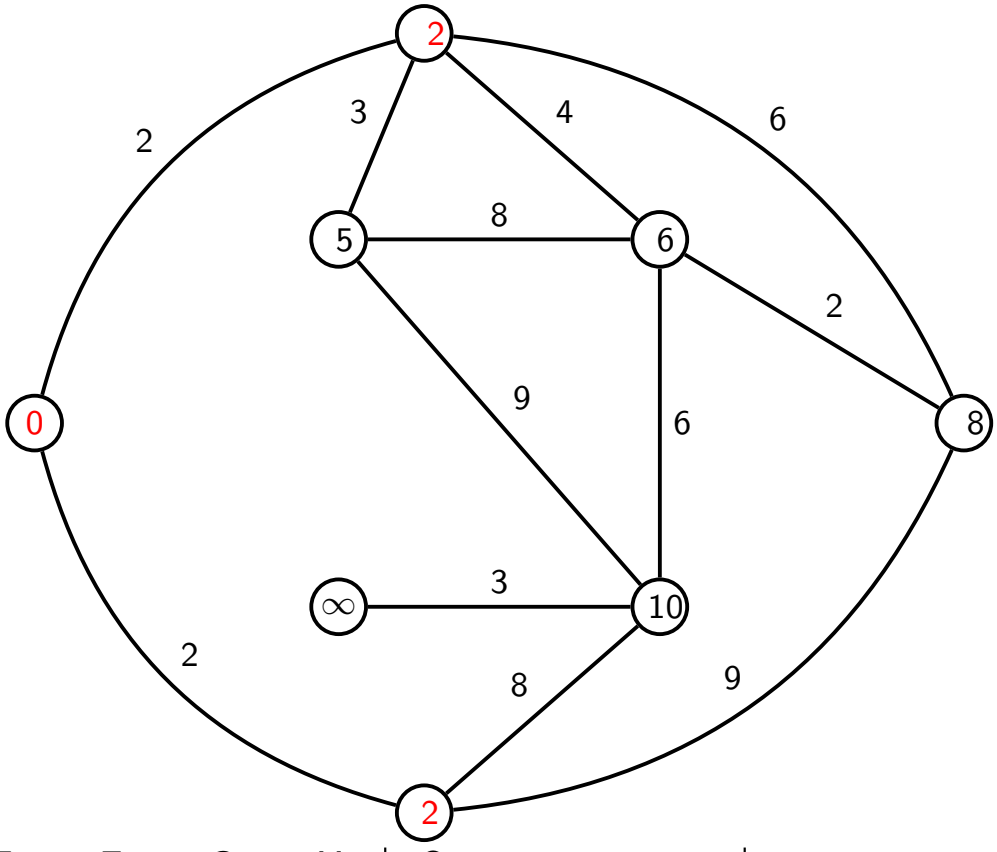




	A	B	C	D	E	F	G	H	Q	u
0	0	∞	∞	∞	∞	∞	∞	∞	A,B,C,D,E,F,G,H	A
1	0	∞	∞	∞	∞	2	2	∞	B,C,D,E,F,G,H	G
2	0	∞	∞	∞	10	2	2	11	B,C,D,E,F,H	F

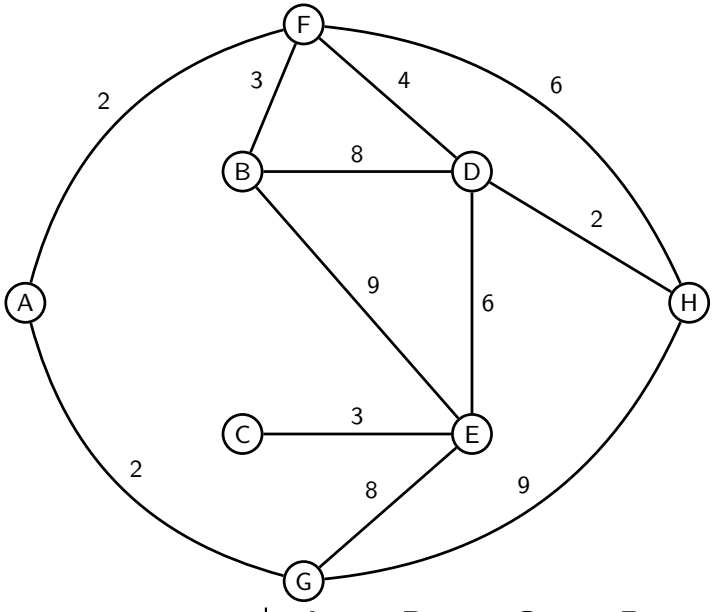
Tracing the algorithm: second example

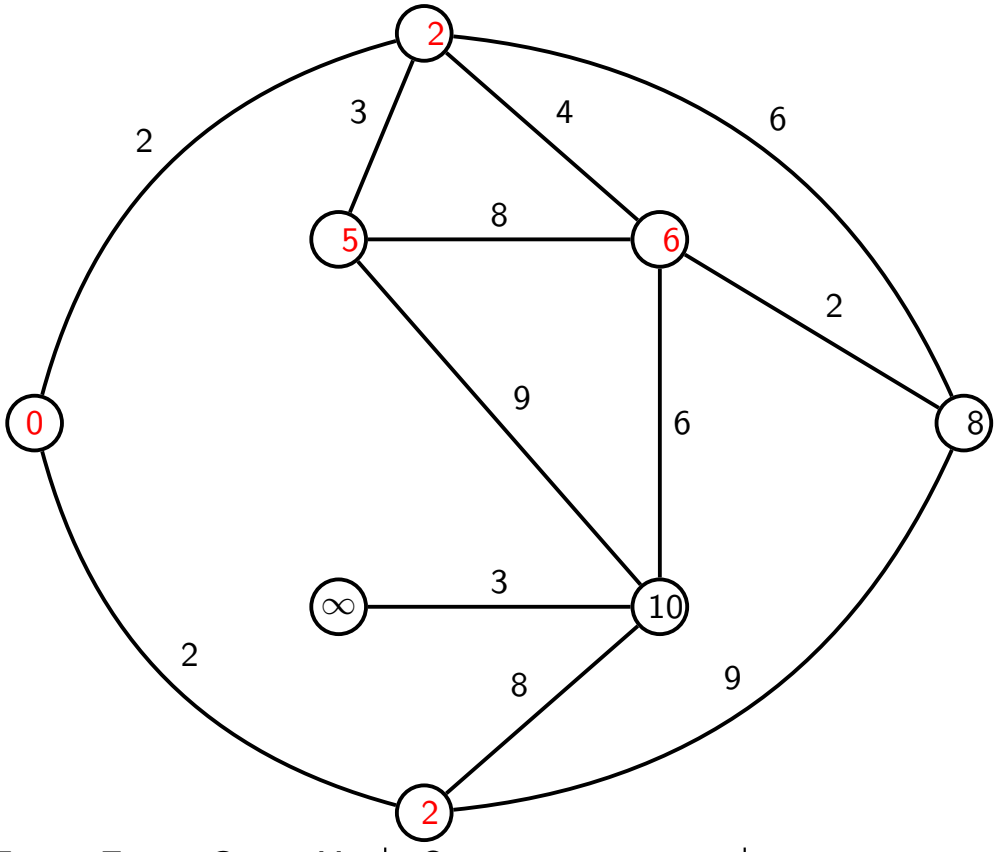




	A	B	C	D	E	F	G	H	Q	u
0	0	∞	∞	∞	∞	∞	∞	∞	A,B,C,D,E,F,G,H	A
1	0	∞	∞	∞	∞	2	2	∞	B,C,D,E,F,G,H	G
2	0	∞	∞	∞	10	2	2	11	B,C,D,E,F,H	F
3	0	5	∞	6	10	2	2	8	B,C,D,E,H	B

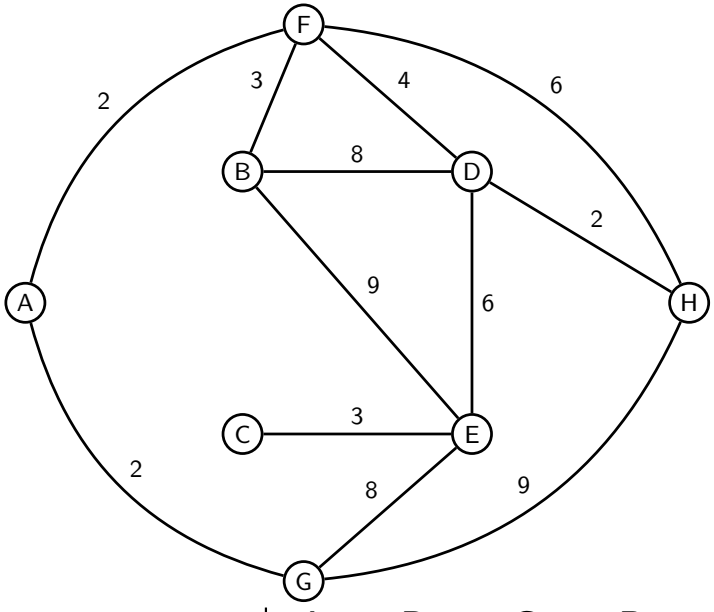
Tracing the algorithm: second example

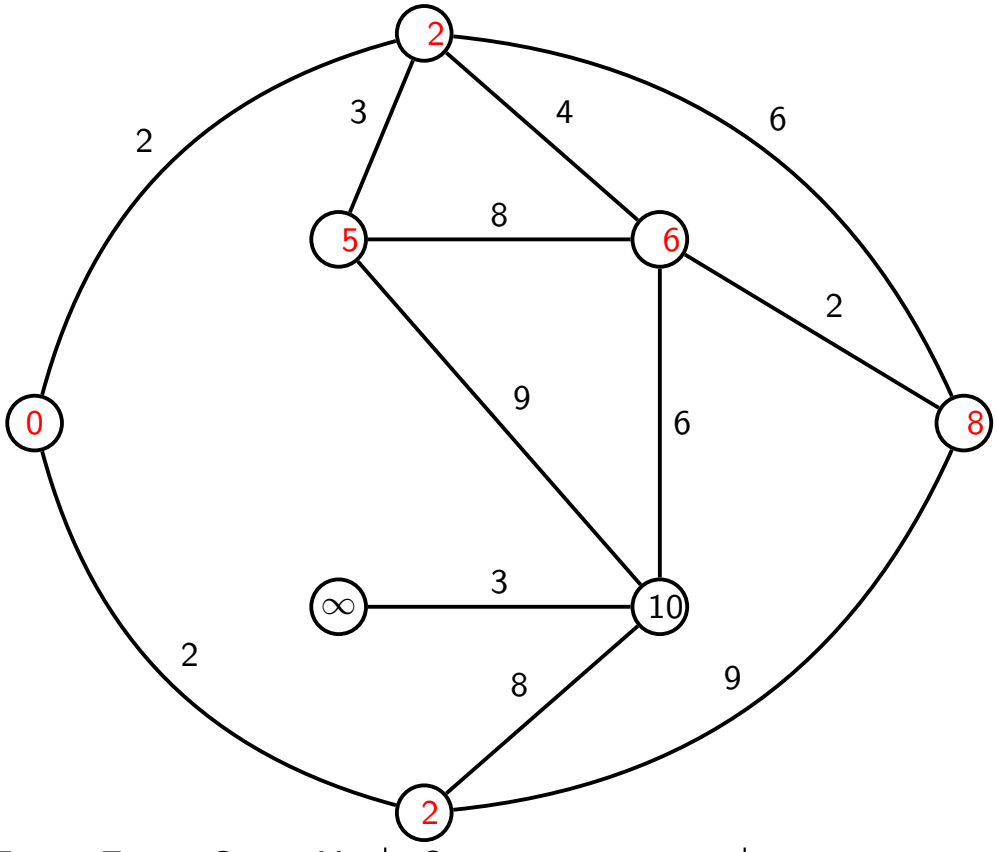




	A	B	C	D	E	F	G	H	Q	u
0	0	∞	∞	∞	∞	∞	∞	∞	A,B,C,D,E,F,G,H	A
1	0	∞	∞	∞	∞	2	2	∞	B,C,D,E,F,G,H	G
2	0	∞	∞	∞	10	2	2	11	B,C,D,E,F,H	F
3	0	5	∞	6	10	2	2	8	B,C,D,E,H	B
4	0	5	∞	6	10	2	2	8	C,D,E,H	D

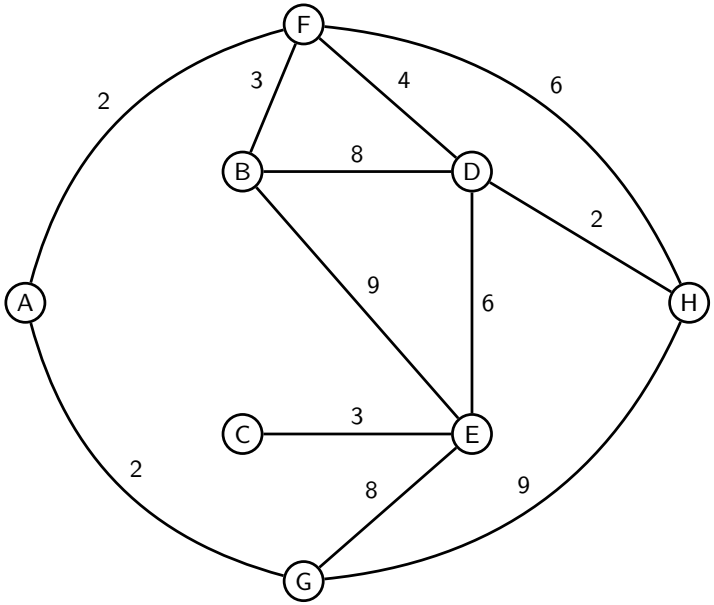
Tracing the algorithm: second example

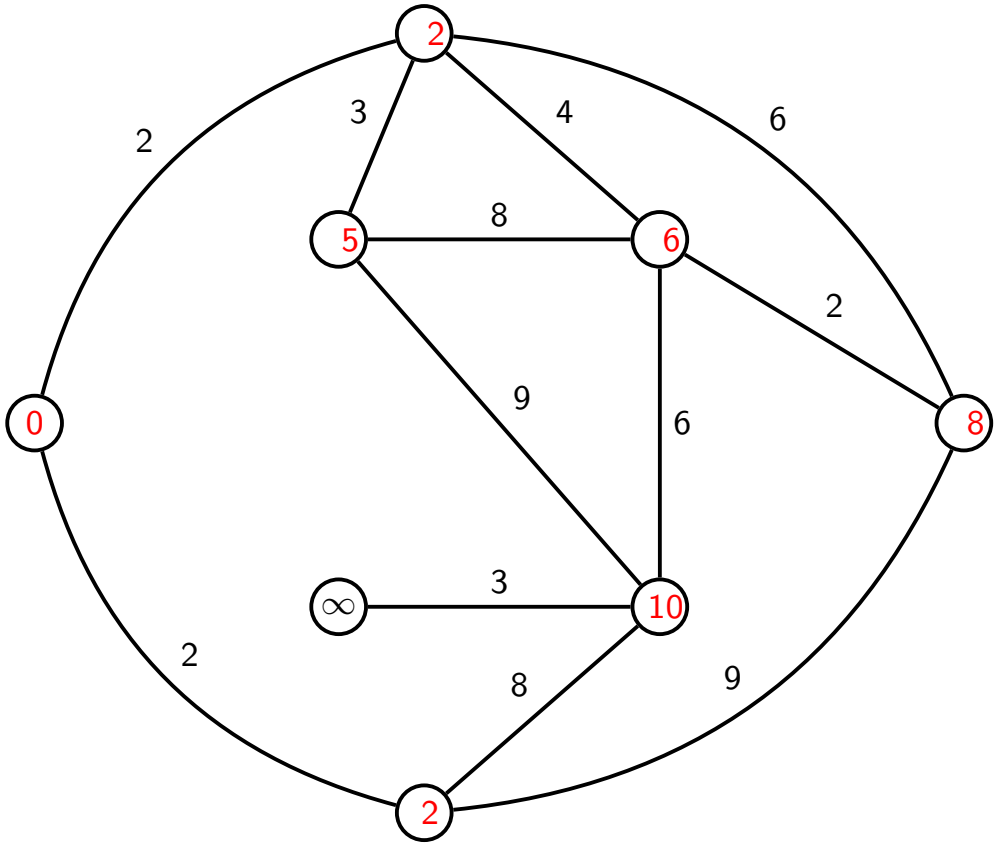




	A	B	C	D	E	F	G	H	Q	u
0	0	∞	∞	∞	∞	∞	∞	∞	A,B,C,D,E,F,G,H	A
1	0	∞	∞	∞	∞	2	2	∞	B,C,D,E,F,G,H	G
2	0	∞	∞	∞	10	2	2	11	B,C,D,E,F,H	F
3	0	5	∞	6	10	2	2	8	B,C,D,E,H	B
4	0	5	∞	6	10	2	2	8	C,D,E,H	D
5	0	5	∞	6	10	2	2	8	C,E, H	H

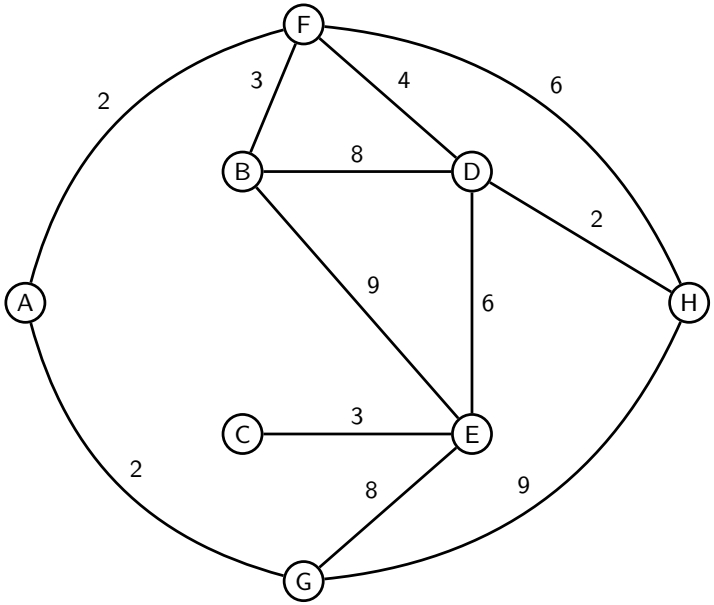
Tracing the algorithm: second example

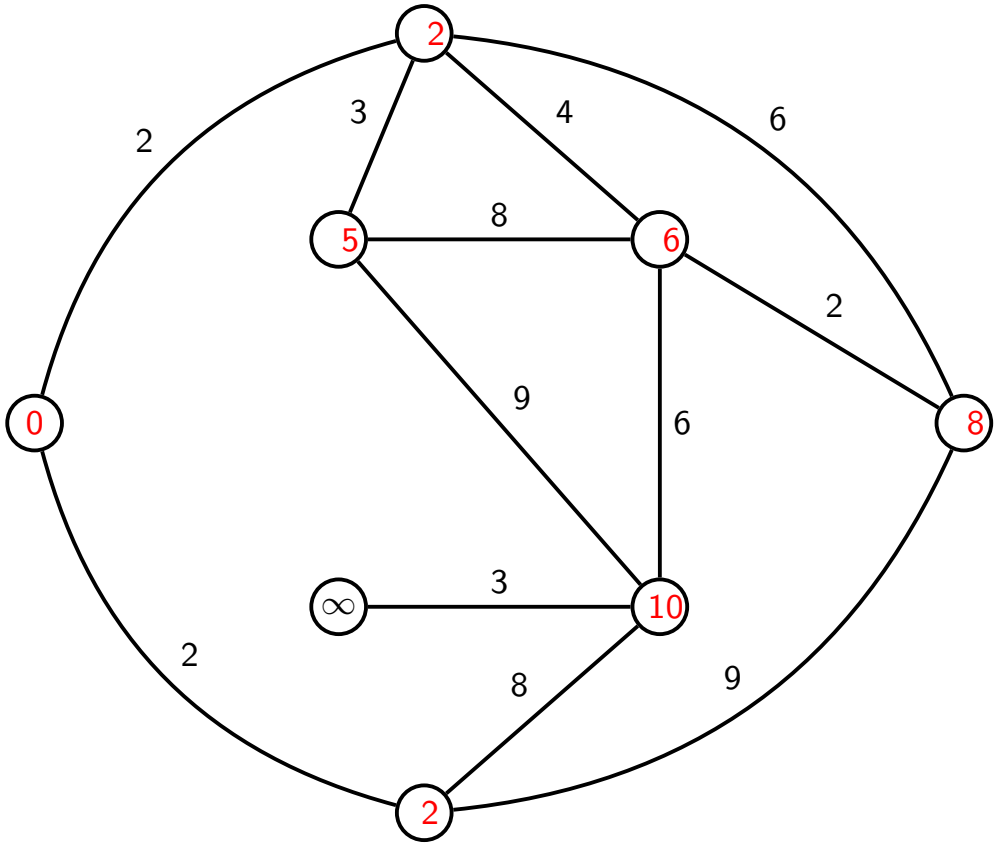




	A	B	C	D	E	F	G	H	Q	u
0	0	∞	∞	∞	∞	∞	∞	∞	A,B,C,D,E,F,G,H	A
1	0	∞	∞	∞	∞	2	2	∞	B,C,D,E,F,G,H	G
2	0	∞	∞	∞	10	2	2	11	B,C,D,E,F,H	F
3	0	5	∞	6	10	2	2	8	B,C,D,E,H	B
4	0	5	∞	6	10	2	2	8	C,D,E,H	D
5	0	5	∞	6	10	2	2	8	C,E, H	H
6	0	5	∞	6	10	2	2	8	C,E	E

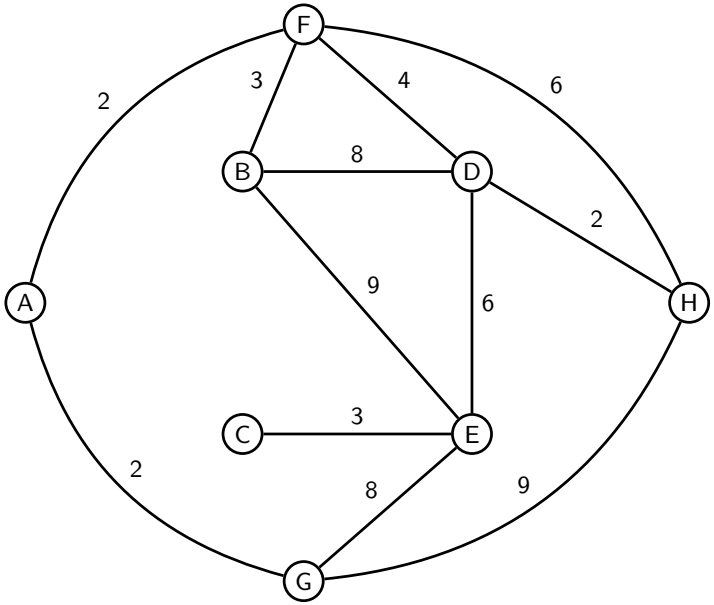
Tracing the algorithm: second example

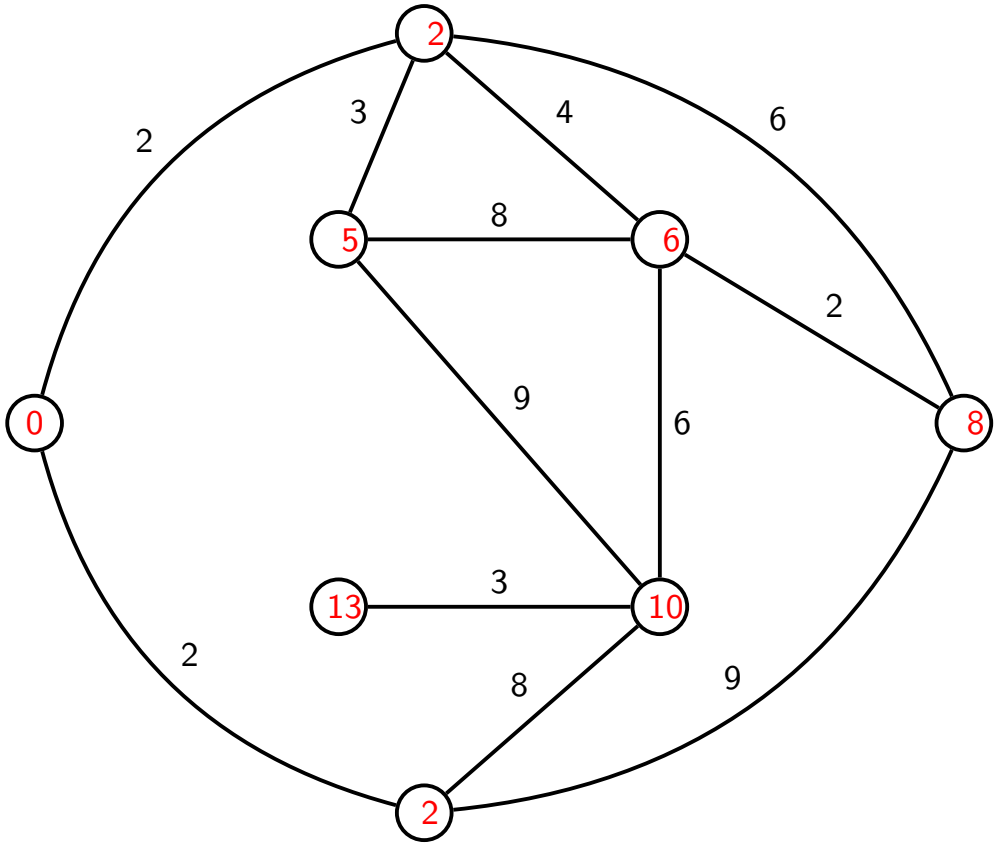




	A	B	C	D	E	F	G	H	Q	u
0	0	∞	∞	∞	∞	∞	∞	∞	A,B,C,D,E,F,G,H	A
1	0	∞	∞	∞	∞	2	2	∞	B,C,D,E,F,G,H	G
2	0	∞	∞	∞	10	2	2	11	B,C,D,E,F,H	F
3	0	5	∞	6	10	2	2	8	B,C,D,E,H	B
4	0	5	∞	6	10	2	2	8	C,D,E,H	D
5	0	5	∞	6	10	2	2	8	C,E, H	H
6	0	5	∞	6	10	2	2	8	C,E	E
7	0	5	13	6	10	2	2	8	C	C

Tracing the algorithm: second example





	A	B	C	D	E	F	G	H	Q	u
0	0	∞	∞	∞	∞	∞	∞	∞	A,B,C,D,E,F,G,H	A
1	0	∞	∞	∞	∞	2	2	∞	B,C,D,E,F,G,H	G
2	0	∞	∞	∞	10	2	2	11	B,C,D,E,F,H	F
3	0	5	∞	6	10	2	2	8	B,C,D,E,H	B
4	0	5	∞	6	10	2	2	8	C,D,E,H	D
5	0	5	∞	6	10	2	2	8	C,E, H	H
6	0	5	∞	6	10	2	2	8	C,E	E
7	0	5	13	6	10	2	2	8	C	C

Dijkstra's algorithm

Overall idea:

1. Base case: Start with Explored = s.
2. Inductive step:

   Optimal substructure: After having computed the shortest path to all vertices in Explored,

   Greedy choice: extend Explored with a v that can be reached using one edge e from some $u \in \mathit{Explored}$ such that

   $$\mathit{dist}(u) + w(e) \text{ is minimized} \tag{3}$$

3. Finish: when Explored = V

Complexity of Dijkstra's algorithm (the good implementation)

- Suppose that n is number of nodes, m is number of edges
- n insertions into Q.
- n calls to the removeMin method on Q.
  - removeMin is $\log n$. Hence this part is $n \log n$
- m calls to the distance checking and updating
  - Updating Adaptable priority queue is $O(\log n)$
- The complexity is $O((n + m) \log n)$ (for good implementation)
- It is also correct to say that it is $O(m \log n)$ because $n < m$.
- There are other variants of implementation that have worse time complexity.

Run Shortest Path Algorithm

- Construct a graph
- Find the shortest paths from s
- Print some paths and distances

```
public class Graph_client {
  public static void main(String[] args) throws Exception {
    //construct a graph from a text file
    Graph G = new Graph(new Scanner(new File("graph1.txt")));
    int s = 0;
    //find all the shortest paths of G starting from node s
    //saves both distances (in distTo) and paths (in edgeTo)
    Graph_SP sp = new Graph_SP(G, s);
    // print the first a few shortest paths
    for (int t = 0; t < 5; t++) {
         if (sp.distTo[t] < Double.POSITIVE_INFINITY) {
        System.out.printf("%d => %d (%.2f)  ", s, t, sp.distTo[t]);
        String path = t + "";
        //print the path to node t
        for (Edge e = sp.edgeTo[t]; e!=null; e=sp.edgeTo[e.from()]){
                path = e.from() + "->(" + e.weight() + ")->" + path;
        }
        System.out.println(path);
      } else
        System.out.printf("no path");
    }
  }
}
```

The dilemma of the Dijkstra's algorithm

Need to implement two operations efficiently:

- `removeMin` (use Priority Queue)
- Find and Update (use `HashTable`)

Algorithm 6: Dijkstra's algorithm

Input: Graph G=(V, E), starting node *s*
Output: Distance *dist* to all remaining nodes

```
1  dist[s] =0;
2  dist[v]=∞ for all other nodes;
3  PriorityQueue Q=nodes in V with their distances;
4  while Q not empty do
5      u =Q.removeMin();
6      for all v ∈ neighbors[u] and v ∈ Q do
7          if dist[v] > dist[u] + w(u, v) then
8              dist[v]=dist[u] + w(u, v);
9              update Q with new distance (put(v, dist(v), or put(dist(v), v) ?)
10         end
11     end
12 end
13 return dist
```

First round: Shortest Path With a Bad (but easier) Data Structure

```
public class Graph_SP {
  public double[] distTo; //record the distances
  public Edge[] edgeTo; //records the edges along the paths
  public Map distTable=new HashMap<Integer, Double>(); //track
      remaining nodes
  public Graph_SP(Graph G, int s) {
    distTo = new double[G.V];  //array of length of number of nodes
    edgeTo = new Edge[G.V];
    //Initially every node has infinitely large distance
    for (int v = 0; v < G.V; v++)
      distTo[v] = Double.POSITIVE_INFINITY;
    distTo[s] = 0.0; // starting node has zero distance

    // relax vertices in order of distance from s
    distTable.put(s, distTo[s]);
    while (!distTable.isEmpty()) {
      int v = removeMin(distTable);
      for (Edge e : G.edgesOf[v])
        relax(e);
      }
    }
```

Relaxation

```
private void relax(Edge e) {
        int v = e.from(), w = e.to();
        if (distTo[w] > distTo[v] + e.weight()) {
                distTo[w] = distTo[v] + e.weight();
                edgeTo[w] = e;
                distTable.put(w, distTo[w]);
        }
}
```

removeMin is slow (in bad implementation)

```
public static int removeMin(Map<Integer, Double> distTable) {
        Iterator it = distTable.entrySet().iterator();
        double minValue = Integer.MAX_VALUE;
        int minKey = -1;
        while (it.hasNext()) {
                Map.Entry pair = (Map.Entry) it.next();

                if (minValue > (Double) pair.getValue()) {
                        minKey = (Integer) pair.getKey();
                        minValue = (Double) pair.getValue();
                }
        }
        distTable.remove(minKey);
        return minKey;
}
```

Time complexity for the naive implementation:

- m edges need to be checked/updated.
  - Check is fast for HashMap (O(1)).
- There are n removeMin operations.
  - the cost of RemoveMin is O(n)
- Hence altogether it is $O(m) + O(n^2)$
- It can be simplified as $O(n^2)$ because $m < n^2$.

The Better One: Heap2540_SP

It is similar to Heap2540, but adapted for the SP problem

- minHeap, not maxHeap. (change `less(..)` to `greater(..)`)
- need to save both the node and the distance (use `distances`)
- need to locate and update quickly (use `index`)

```
public class Heap2540_SP {
        int[] heap;
        int[] index;
        Double[] distances;
        int n = 0; // actual heap size.
        int CAPACITY = 1000001; // array size

        public Heap2540_SP() {
                heap = new int[CAPACITY];
                index = new int[CAPACITY];
                distances = new Double[CAPACITY];
                for (int i = 0; i < CAPACITY; i++)
                        index[i] = -1;
        }
```

The Better One: Heap2540_SP

```
public int removeMin()
        int min = heap[1];
        swap(1, n);
        n--;
        sink(1);
        index[min] = -1;
        distances[min] = null;
        return min;

public void insert(int node, double dist)
        n++;
        heap[n] = node;
        index[node] = n;
        distances[node] = dist;
        swim(n);

private void swim(int k)
        while (k > 1 && greater(k / 2, k)) {
                swap(k, k / 2);
                k = k / 2;
        }

public void put(int node, double dist)
        if (index[node] != -1) {
                distances[node] = dist;
                swim(index[node]);
        } else
                insert(node, dist);
```

The Better One: Heap2540_SP

```
        private void swap(int i, int j) {
                Integer temp = heap[i];
                heap[i] = heap[j];
                heap[j] = temp;
                // add  missing code here
        }

        private void sink(int k) {
                while (2 * k <= n) {
                        int j = 2 * k;
                        if (j < n && greater(j, j + 1))
                                j++;
                        if (!greater(k, j))
                                break;
                        swap(k, j);
                        k = j;
                }
        }

// check whether the distance of the node in heap[i] is greater
     than the distance of node in heap[j]
private boolean greater(int i, int j) {
 //heap[i]>heap[j] is not the correct answer. it compares the node
      IDs, which is meaningless. We need to compare their distances.
}
}
```

Lessons learned

- Four levels of programmers
  1. Can write the code (word counts with O(nmm) time complexity)
  2. Can use API correctly (avoid using `get(i)`, obtain $O(nm)$ time complexity)
  3. Can select better data structure (use `hashtable`)
  4. Can modify the API (modify the heap in Dijkstra algorithm)
- Why not justing making PQ adaptable?
  - Extra cost (need the index array)
  - Normally not needed

Takeaways

- Graph definition, G=(V,E). Basic properties. Path. SCC, WCC.
- Graph representations. Adjacency Matrix, Edge list, Adjacency List.
- Graph traversal. DFS. BFS.
- Shortest path. Dijkstra's algorithm. Greedy algorithm.
- Data structures for the Dijkstra's algorithm
- Three steps to understand Dijkstra algorithm:
  - BFS: Shortest path algorithm for unweight graph
  - GraphSP: shortest path without heap
  - Dijkstra algorithm with adaptable heap/priority queue
- Readings:
  - P611-642. P 653-661. Goodrich et al.
  - Algorithms, 4th Edition, By Sedgewick et al. (clickable).